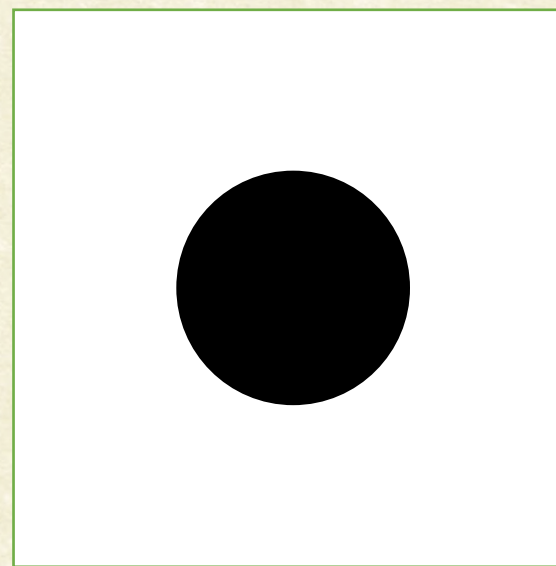
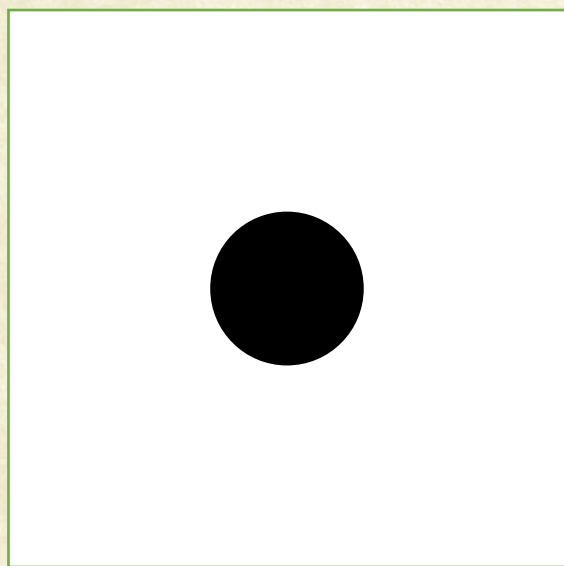




CS7.404: Digital Image Processing

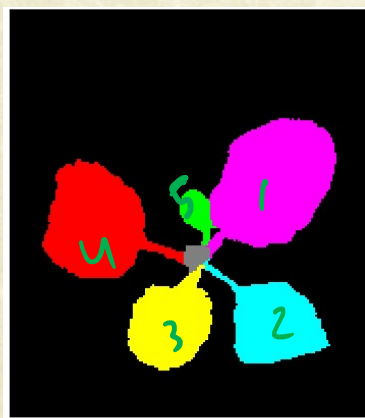
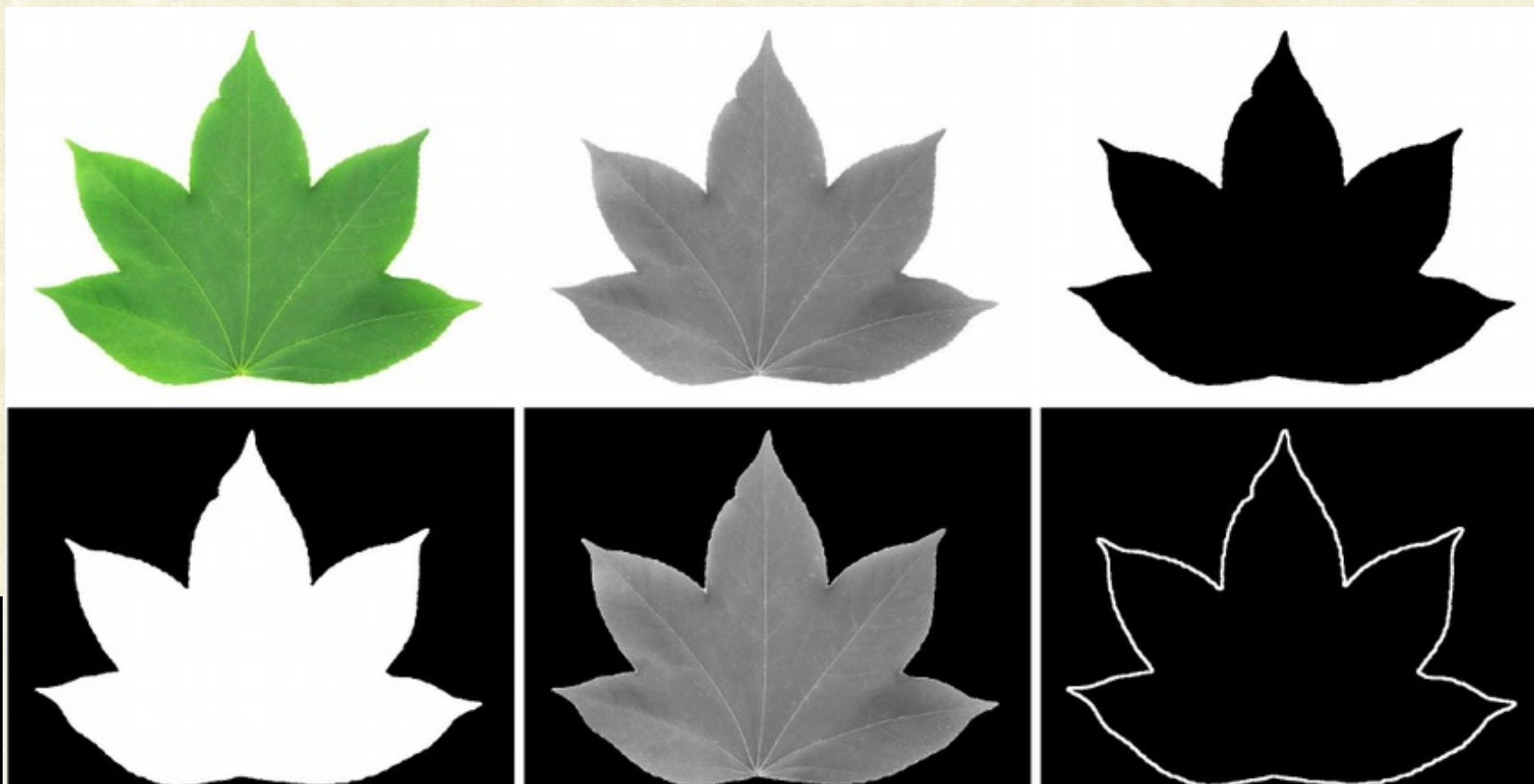
Monsoon 2025: Morphological Image Processing 1



Anoop M. Namboodiri
Biometrics and Secure ID Lab, CVIT,
IIIT Hyderabad



Plant Phenotyping





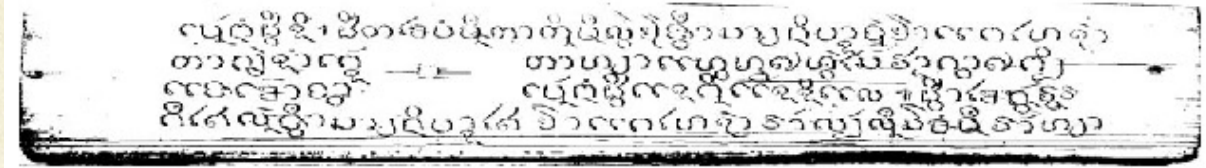
Cleaning Binarized Text



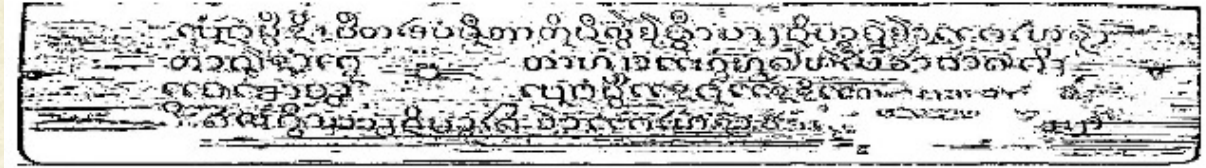
a) RGB image



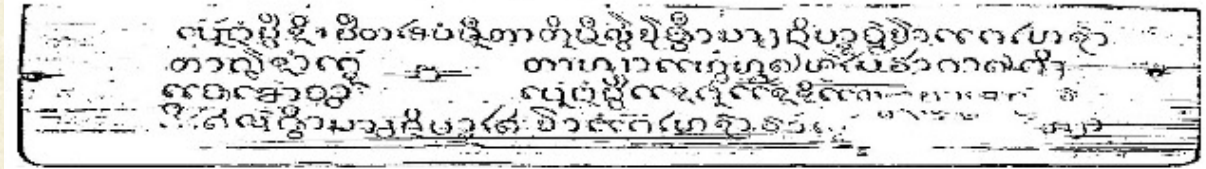
b) Noise reduction image



c) Binary image by Otsu's algorithm



d) Binary image by Niblack's algorithm



e) Binary image by Sauvola's algorithm

Figure 2. Samples of palm leaf images



Other Applications



Minutiae
Extraction



Background
Subtraction



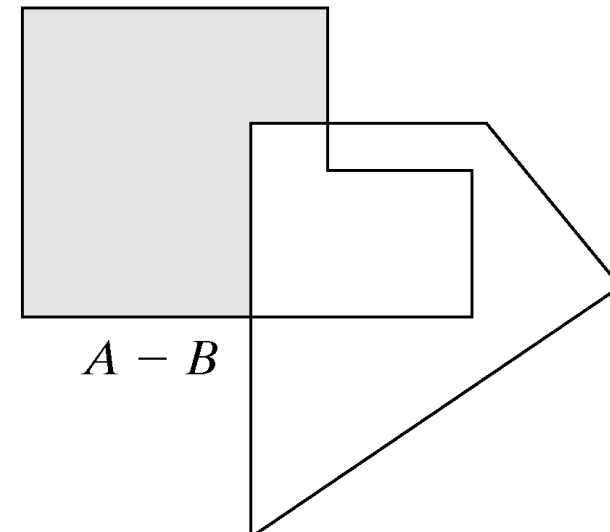
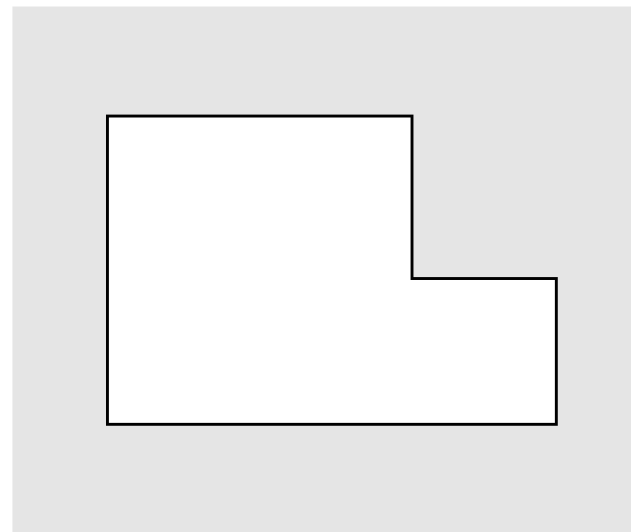
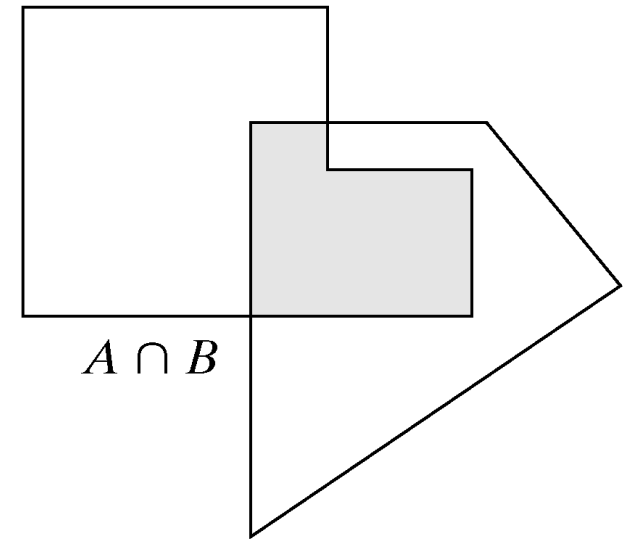
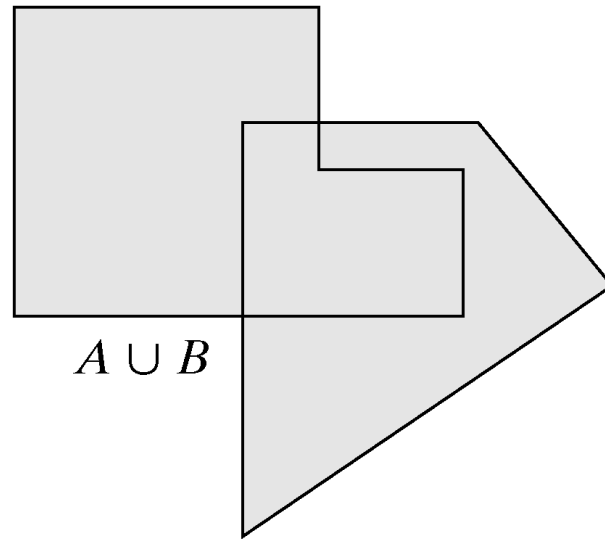
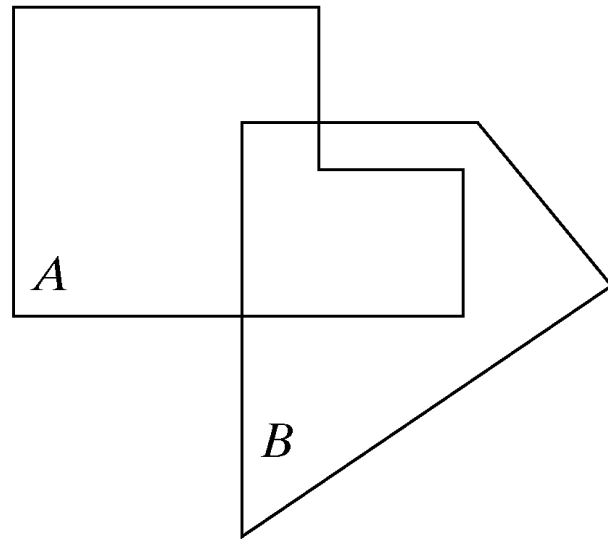
Image/Objects as a Set of Pixels

- Basic idea:
 - Object/Region = Set of pixels (coordinates)
- Each pixel is labelled:
 - 0 = background
 - 1 = foreground





Basic Operations on Shapes/Sets





Structuring Element

Box

1	1	1
1	1	1
1	1	1

3x3

1	1	1	1	1
1	1	1	1	1
1	1	1	1	1
1	1	1	1	1
1	1	1	1	1

5x5

1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

15x15

Disc

0	1	0
1	1	1
0	1	0

0	1	1	1	0
1	1	1	1	1
1	1	1	1	1
1	1	1	1	1
0	1	1	1	0

0	0	0	0	0	1	1	1	1	1	1	0	0	0	0
0	0	0	1	1	1	1	1	1	1	1	1	0	0	0
0	0	1	1	1	1	1	1	1	1	1	1	1	0	0
0	1	1	1	1	1	1	1	1	1	1	1	1	1	0
0	1	1	1	1	1	1	1	1	1	1	1	1	1	0
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
0	1	1	1	1	1	1	1	1	1	1	1	1	1	0
0	1	1	1	1	1	1	1	1	1	1	1	1	1	0
0	0	1	1	1	1	1	1	1	1	1	1	1	0	0
0	0	0	1	1	1	1	1	1	1	1	1	1	0	0
0	0	0	0	0	1	1	1	1	1	0	0	0	0	0



Structuring Element (Kernel)

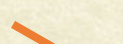
- Can have varying sizes
- Have an origin
- Usually, element values are 0, 1 or none(!)
 - For thinning, other values are possible
- Empty spots in the Structuring Elements (SEs) are nones (*don't cares*)

Box



1	1	1
1	1	1
1	1	1

Disc

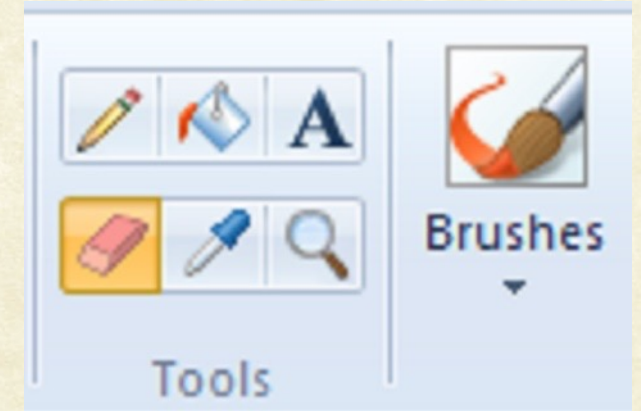


	1	
1	1	1
	1	

		1	1	1		
	1	1	1	1	1	
1	1	1	1	1	1	1
1	1	1	1	1	1	1
1	1	1	1	1	1	1
	1	1	1	1	1	
		1	1	1		

1	1	
1	0	
1		0

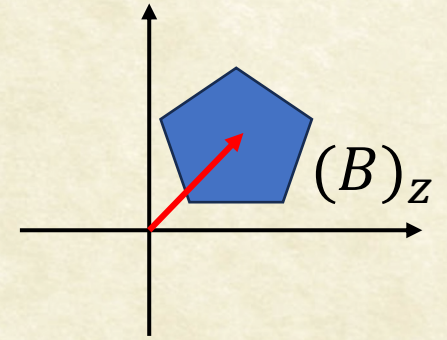
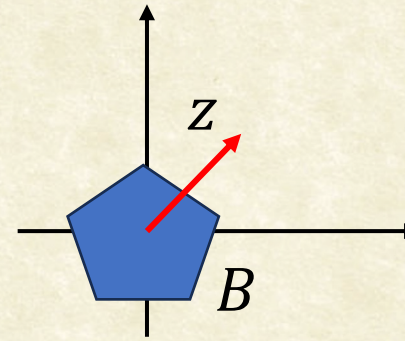
1	1	1
1	10	1
1	1	1



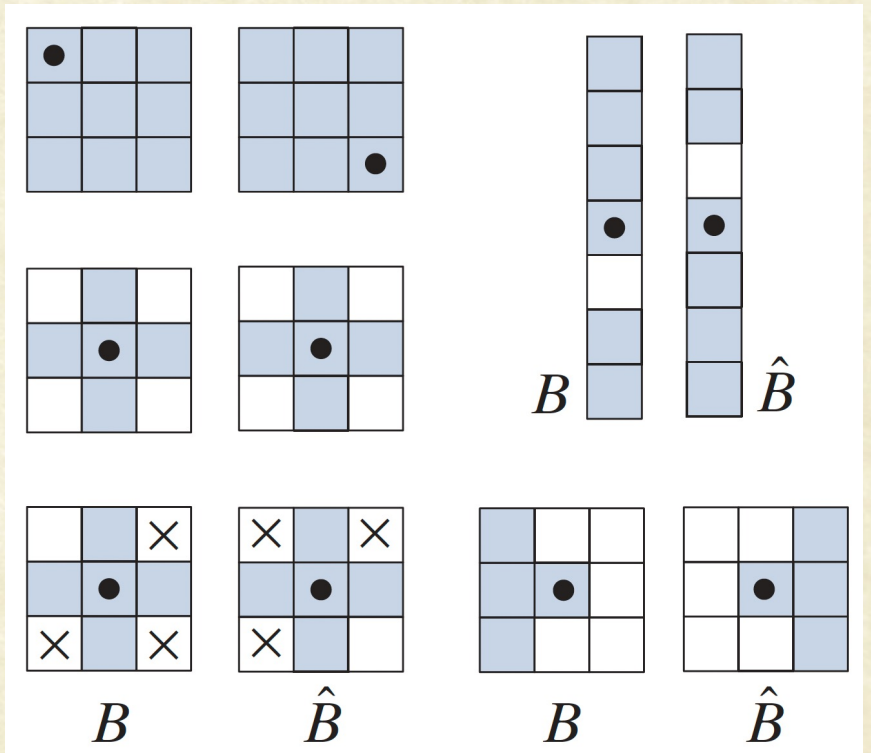


Operations on Structuring Elements

- $(B)_z = \{c \mid c = b + z, \text{ for } b \in B\}$
 - Shift B by z.

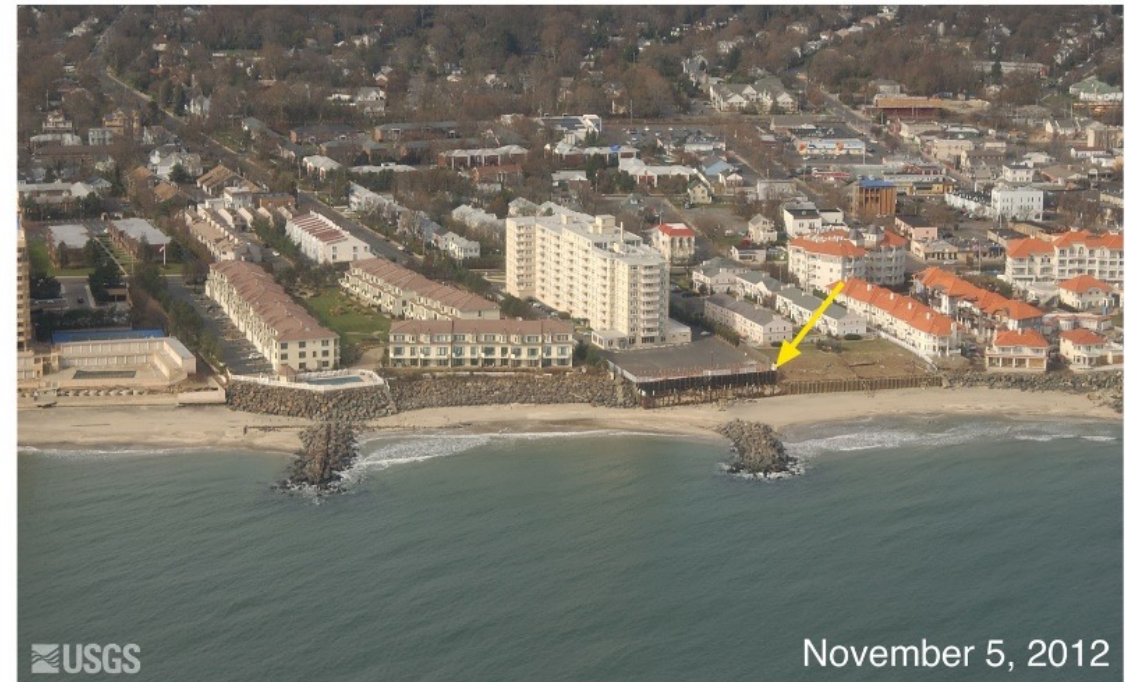


- $\hat{B} = \{w \mid w = -b, b \in B\}$
 - Mirror-Reflect B in both axes





Erosion





Erosion

MASK

MASK

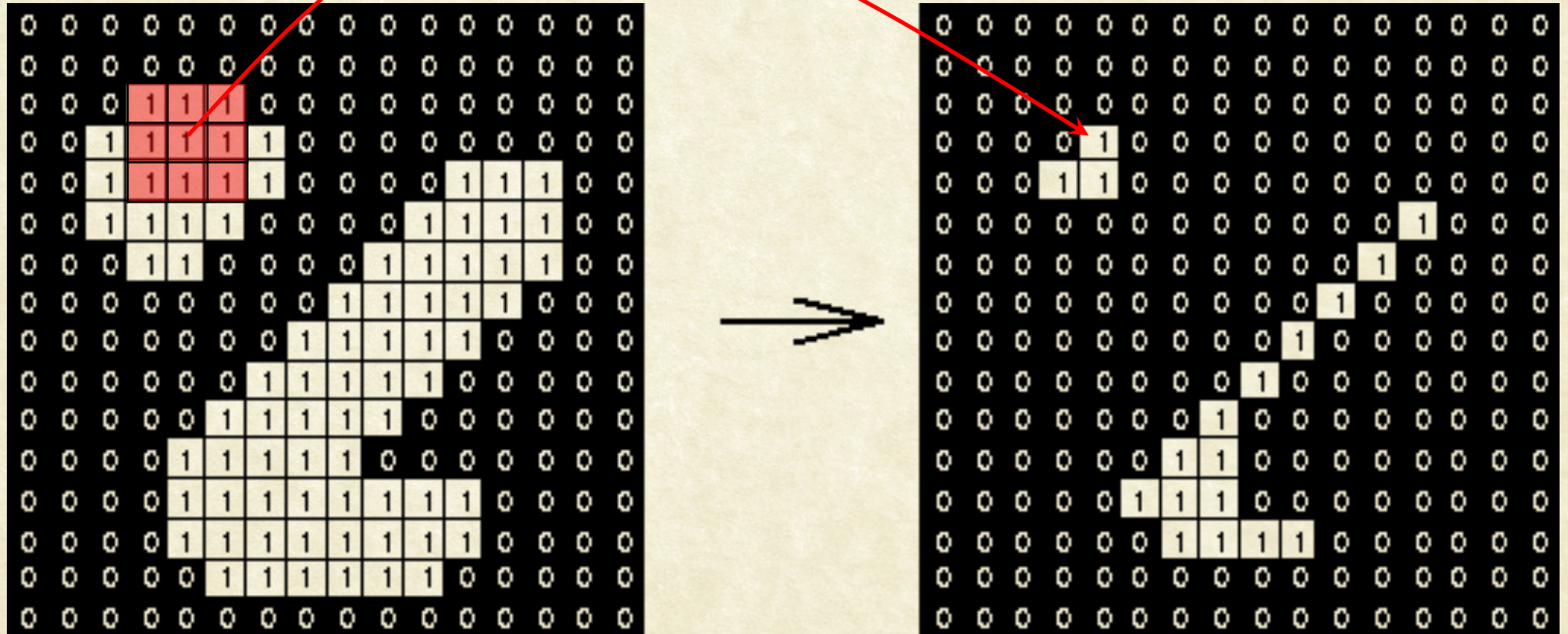




Erosion: Effect

1	1	1
1	1	1
1	1	1

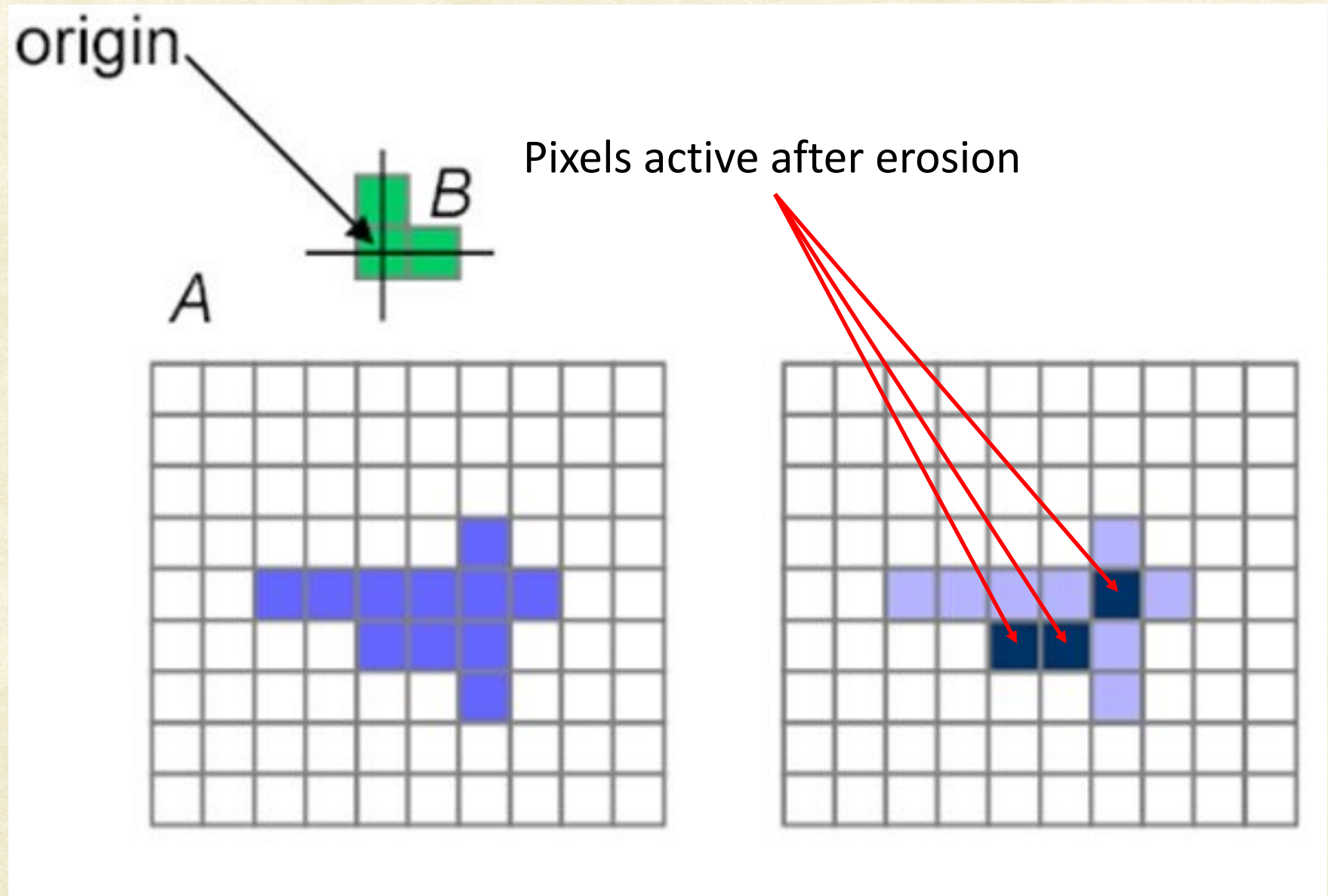
If, for a particular location of Structuring Element (SE) origin, SE lies **fully within the region**, retain the location, else set to 0





SEs Operate w.r.t. an origin

$$A \ominus B = \{z | (B)_z \subseteq A\}$$





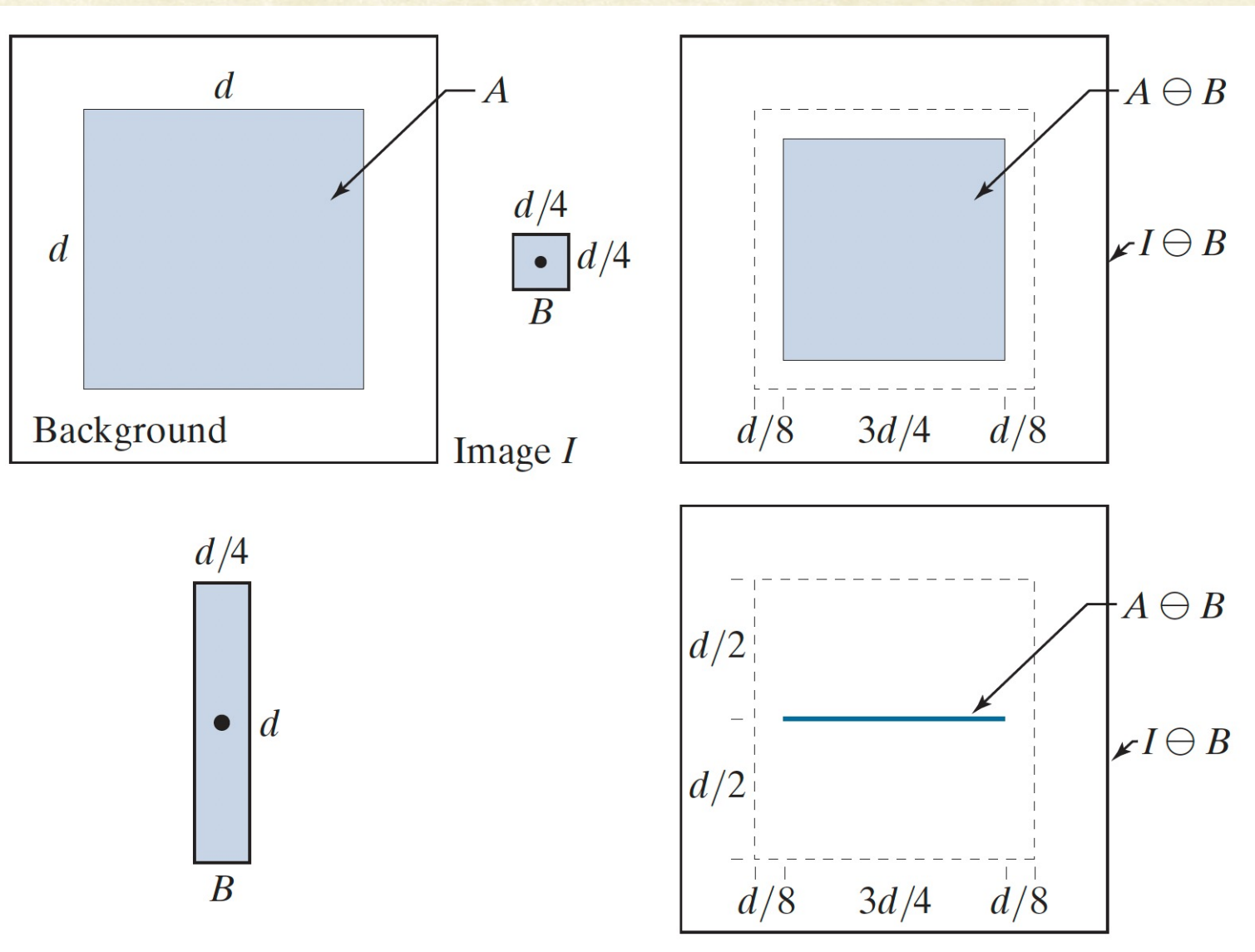
Erosion: Effect of Structuring Element

$$A \ominus B = \{z | (B)_z \subseteq A\}$$

The shape of SE decides the directions of removal of foreground pixels

Alternately:

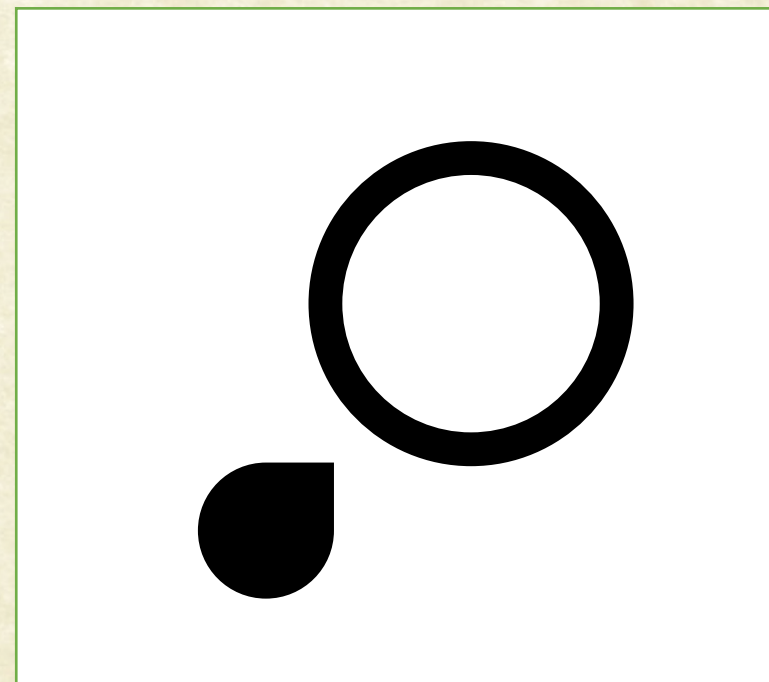
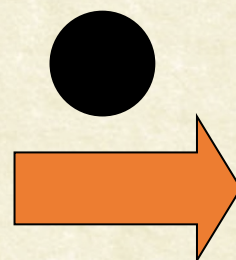
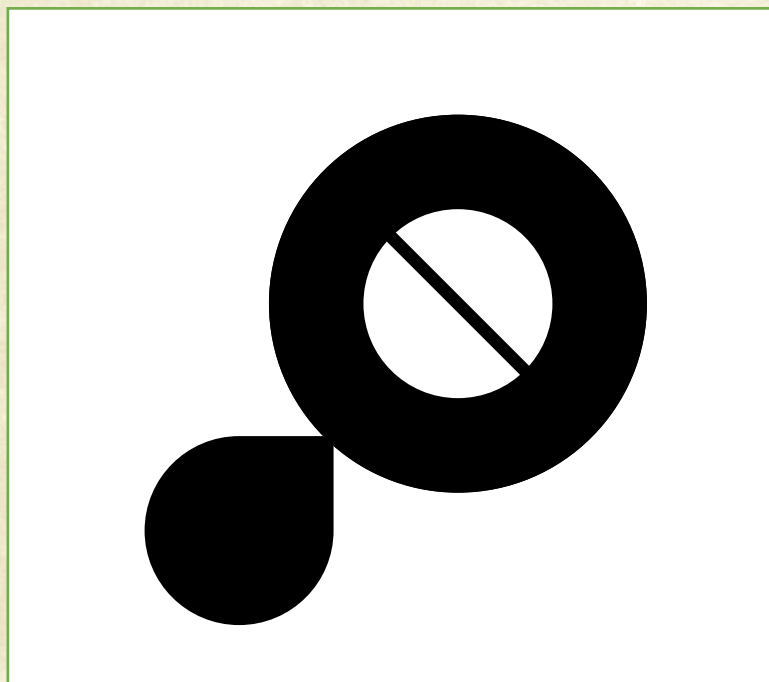
$$A \ominus B = \{z | (B)_z \cap A^c = \emptyset\}$$





Erosion

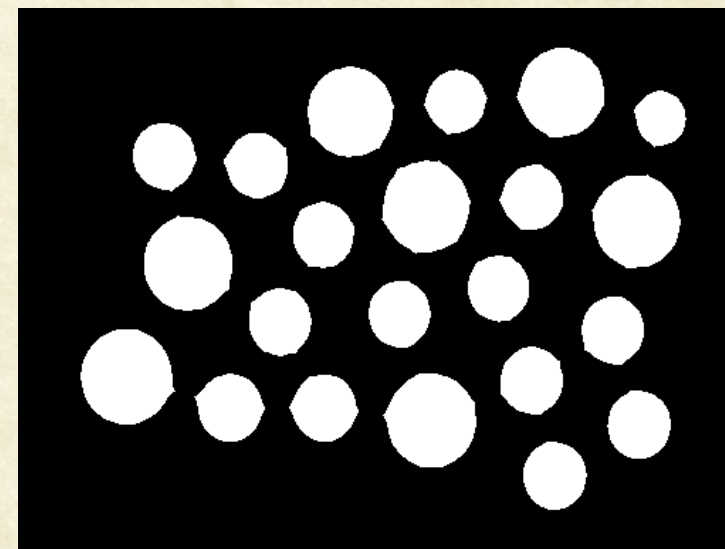
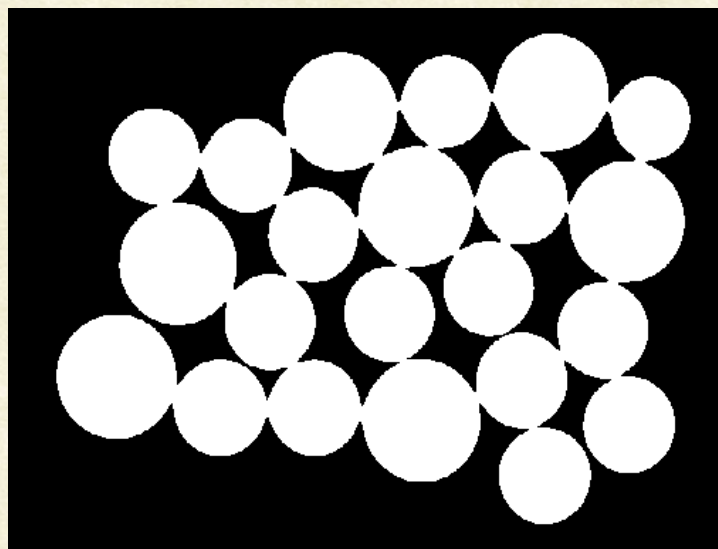
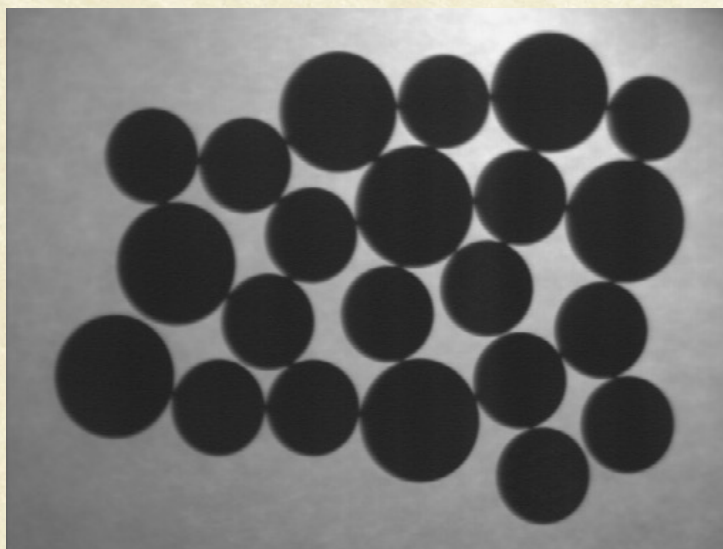
- Shrinks foreground objects
- Thin features are removed
- Touching objects are separated
- NOTE: Multiple iterations of erosion





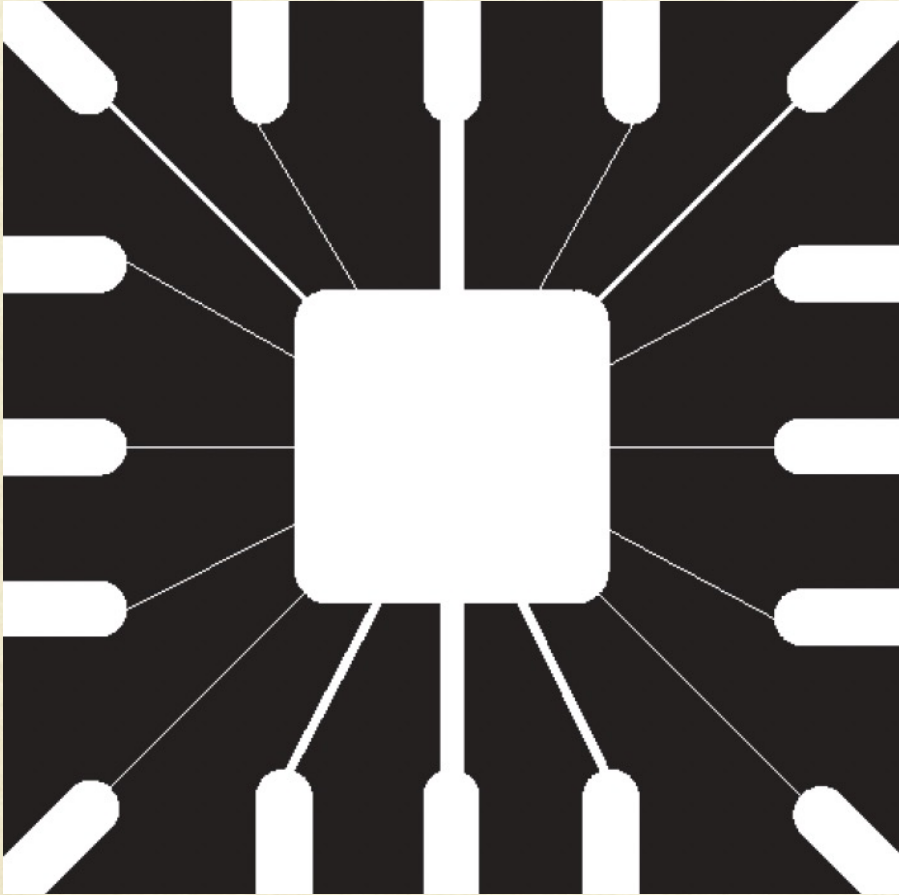
Example: Counting coins

- Difficult because they touch each other!
- Solution: Binarization and Erosion separates them!



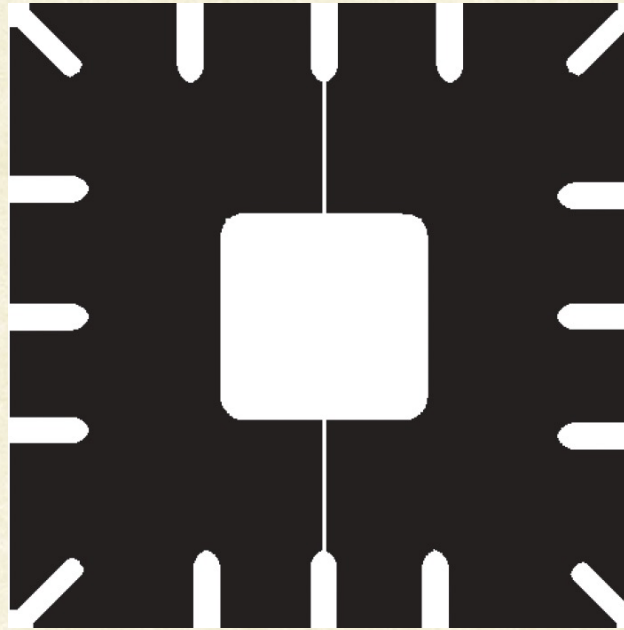


Erosion: Example



486 x 486 image

From: Digital Image Processing, Gonzalez, & Woods

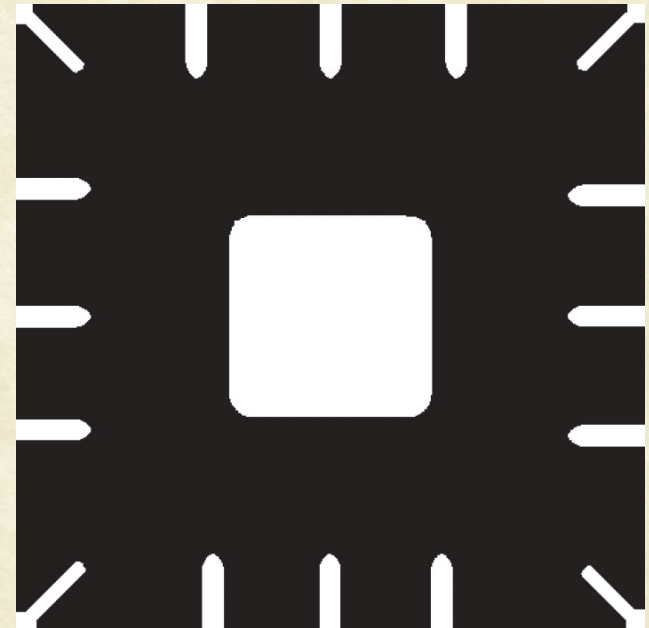
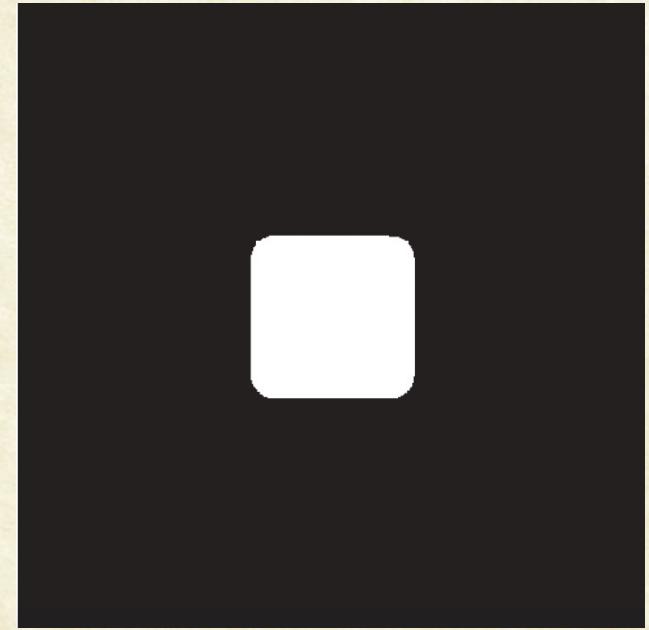


Erosion by square SE:

11 x 11

45 x 45

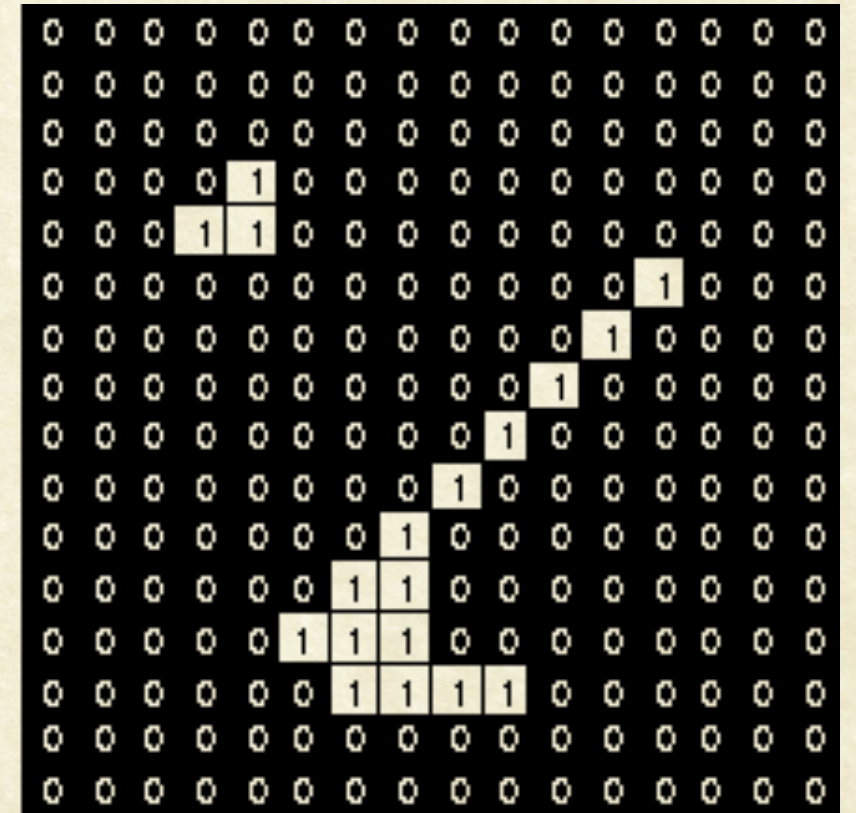
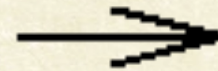
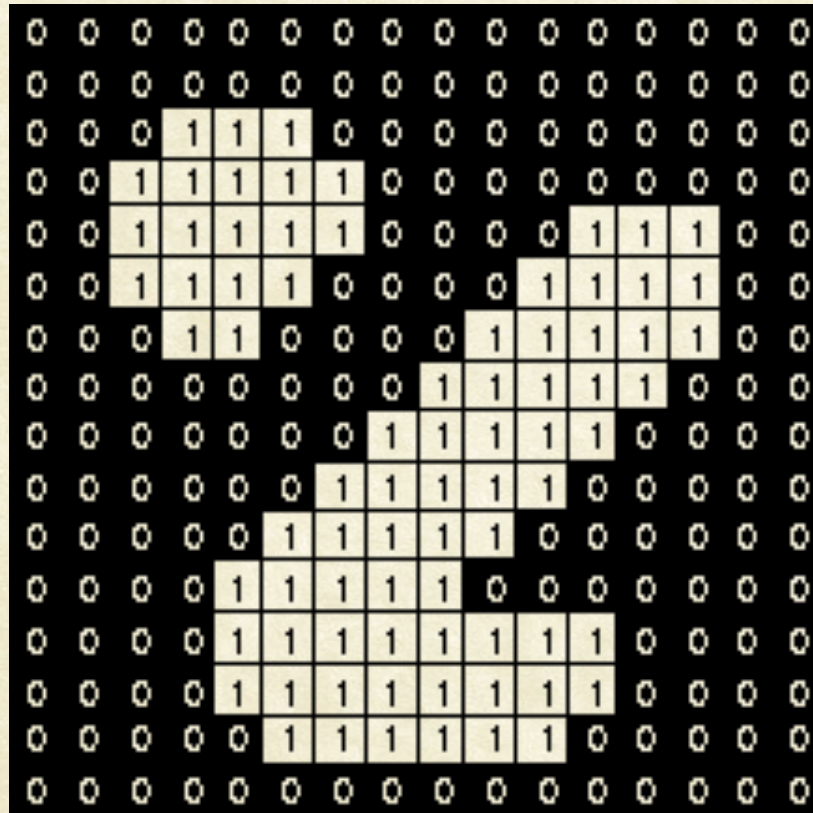
15 x 15





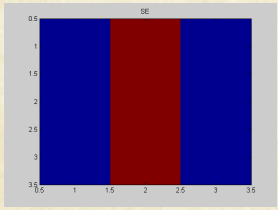
Erosion operation and **min** filter

1	1	1
1	1	1
1	1	1

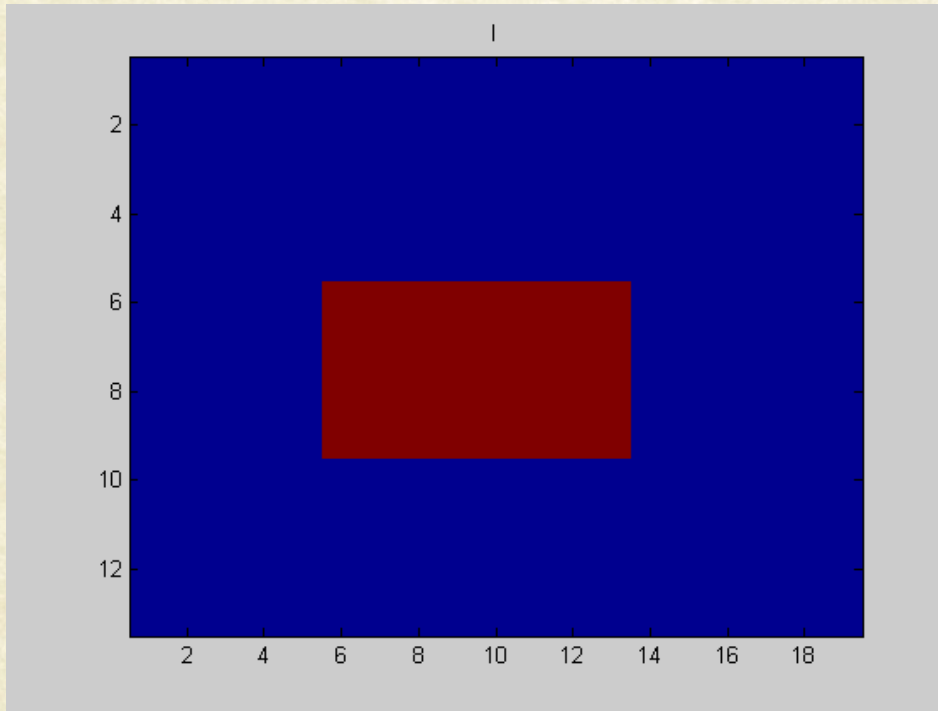




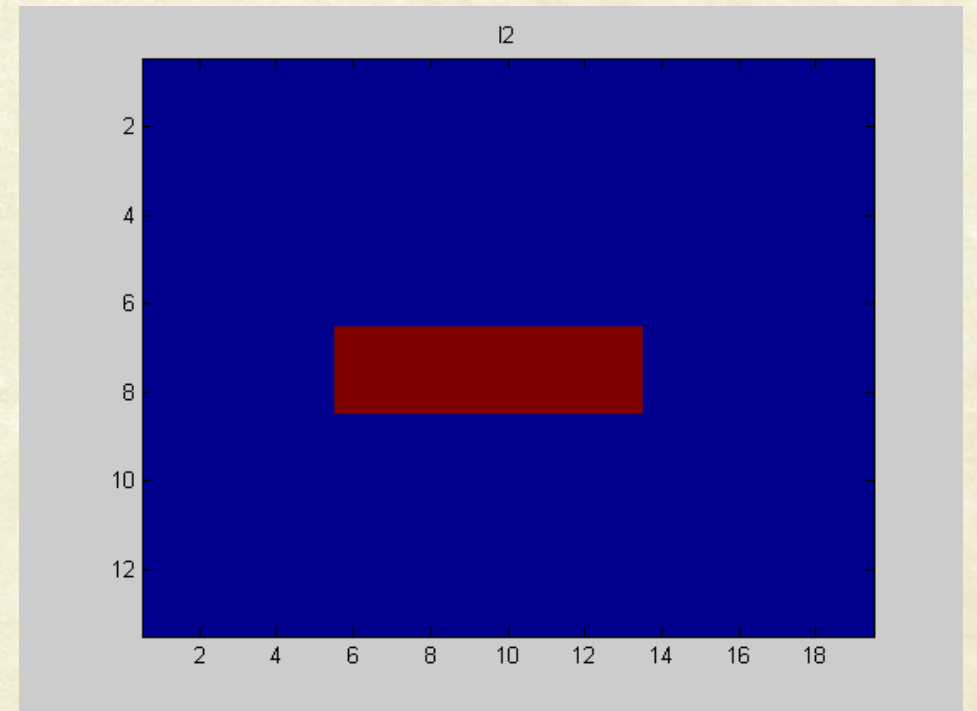
(MATLAB) code



$SE = 3 \times 3$



I2



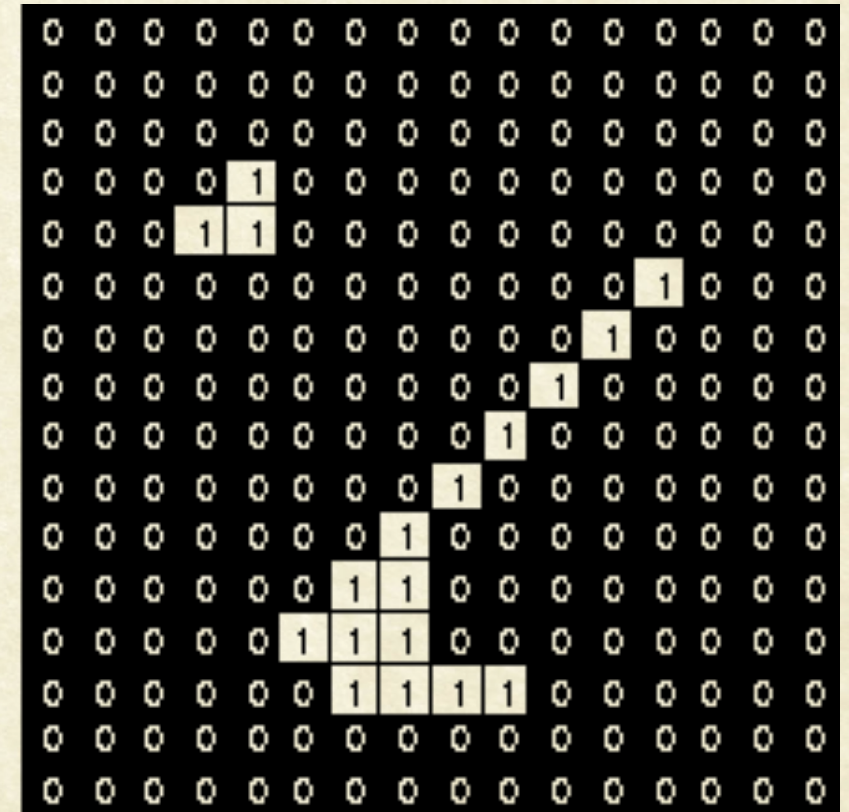
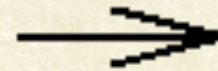
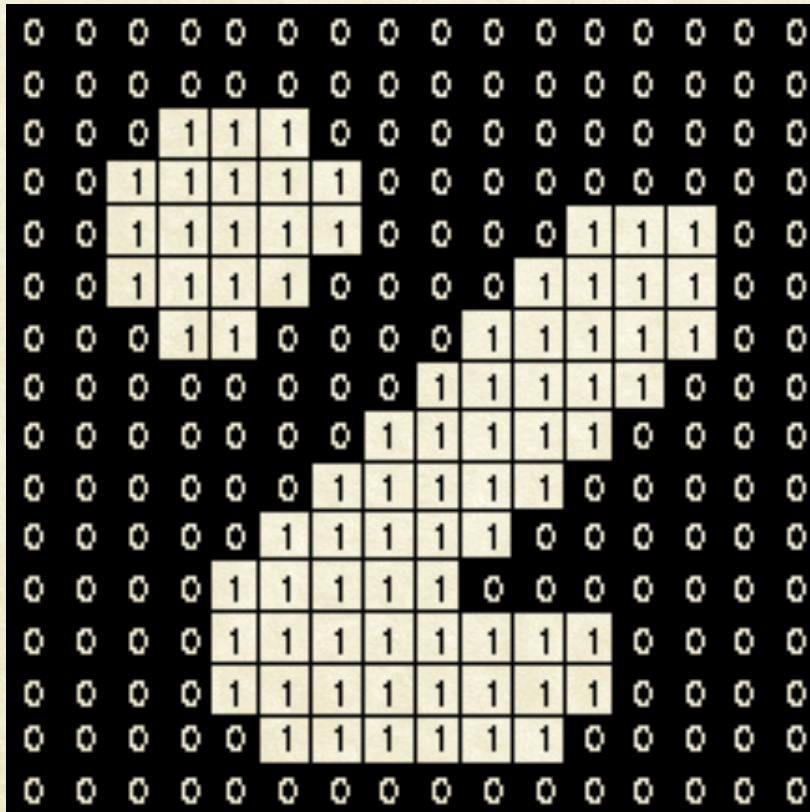
$I3 = \text{imerode}(I2, SE);$



Erosion

1	1	1
1	1	1
1	1	1

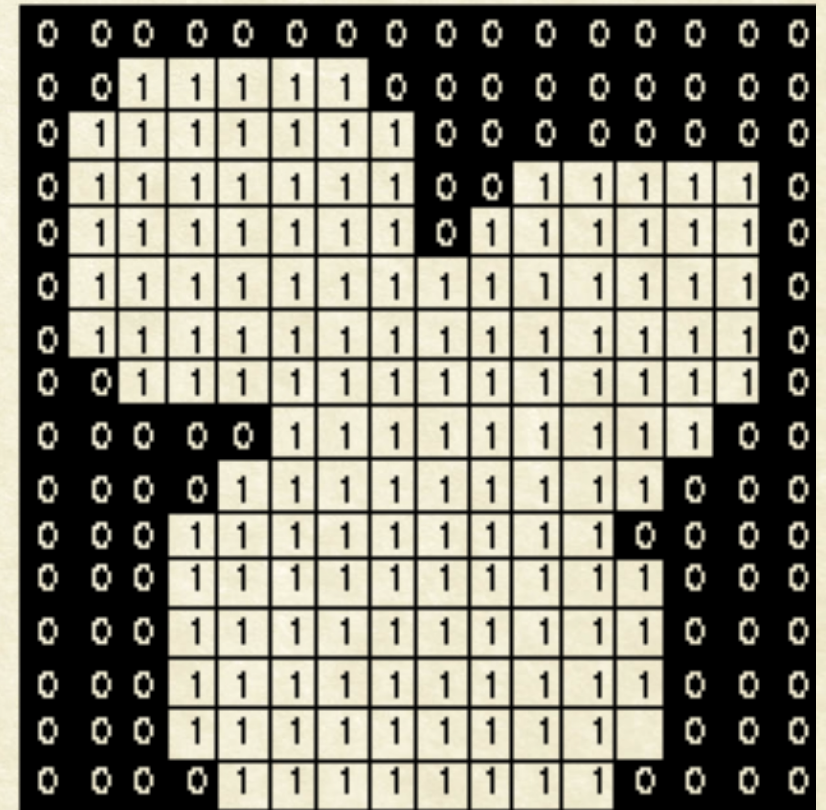
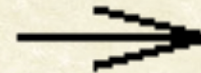
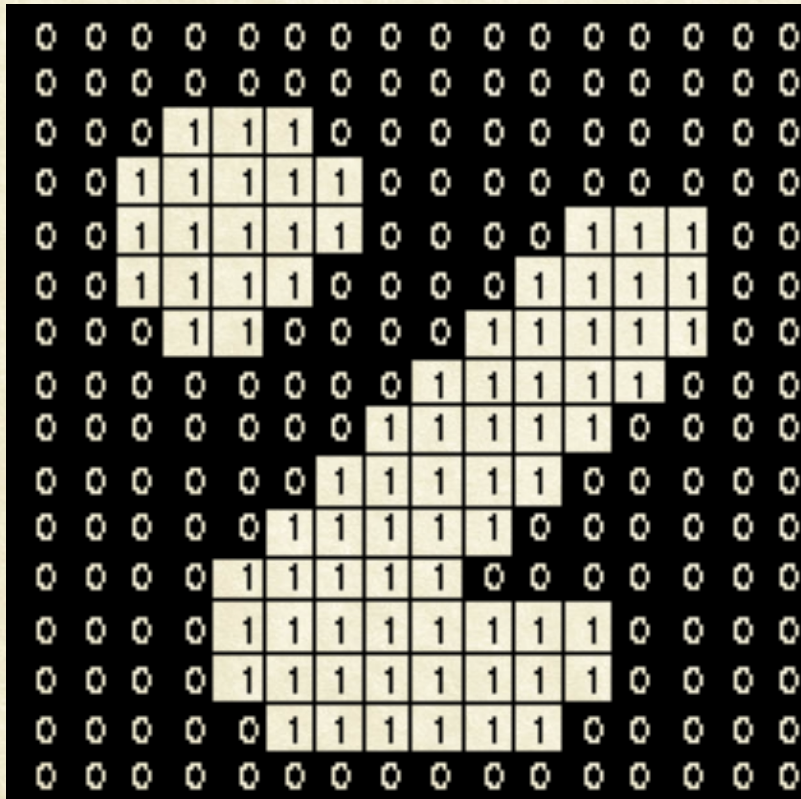
- Simple application of **pattern matching**





Dilation operation (max filter)

1	1	1
1	1	1
1	1	1





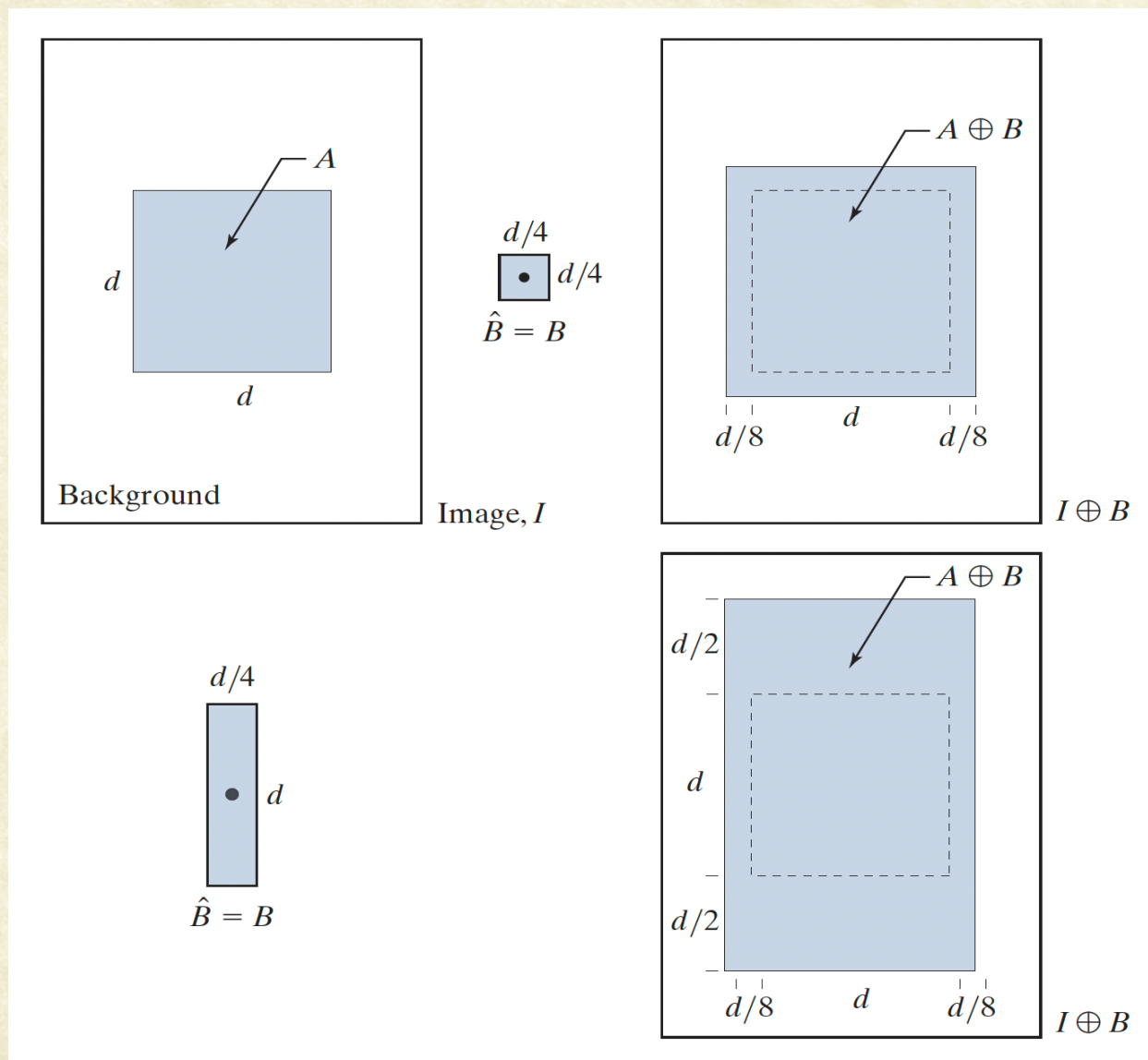
Dilation: Effect of Structuring Element

$$A \oplus B = \{z \mid [(\hat{B})_z \cap A] \neq \emptyset\}$$

The shape of SE
decides the directions
of addition of
foreground pixels

Alternately:

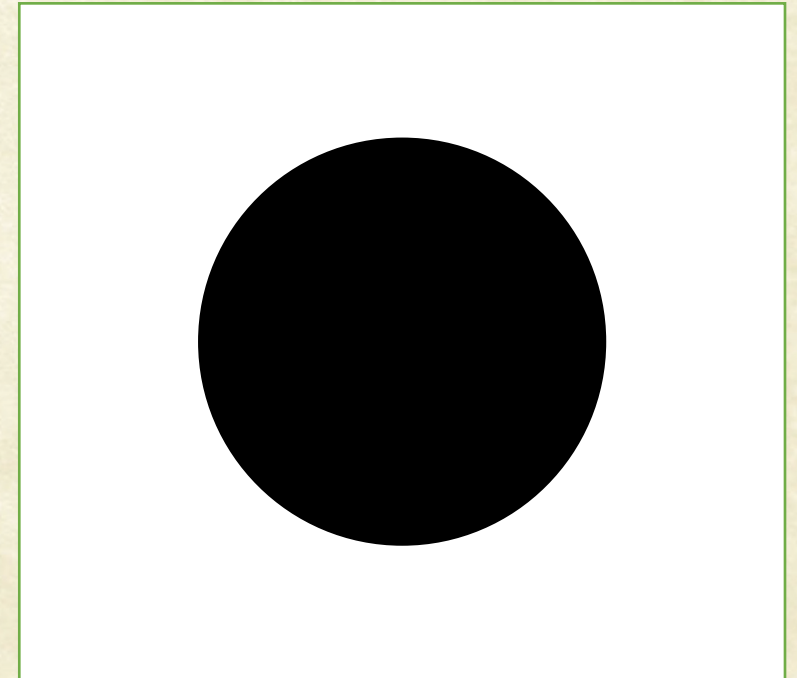
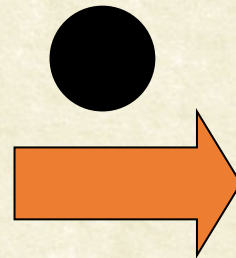
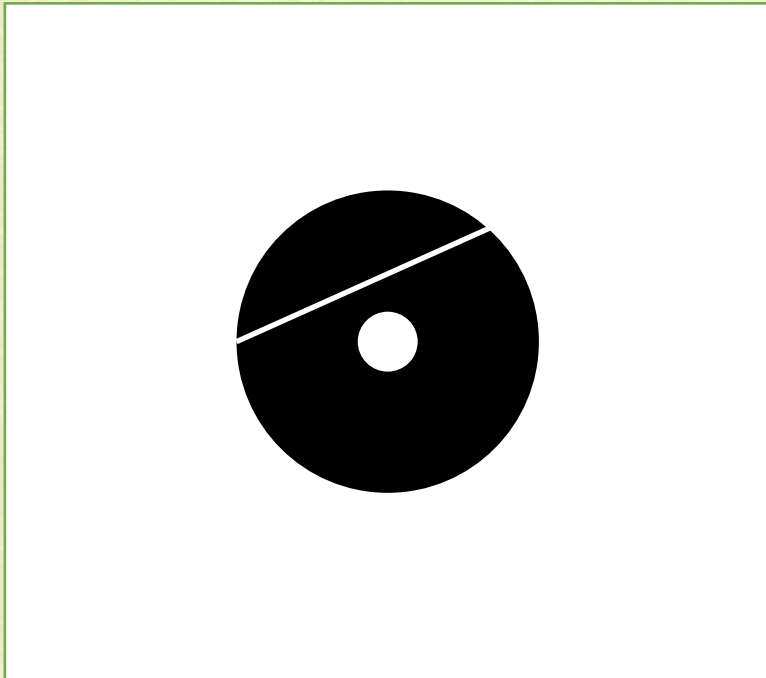
$$A \oplus B = \{z \mid [(\hat{B})_z \cap A] \subseteq A\}$$





Dilation

- Expands foreground objects
- Foreground holes are shrunk / removed
- Broken objects are connected
- NOTE: Multiple iterations of dilation





Dilation: Application

Historically, certain computer programs were written using only two digits rather than four to define the applicable year. Accordingly, the company's software may recognize a date using "00" as 1900 rather than the year 2000.

00

1	1	1
1	1	1
1	1	1



Historically, certain computer programs were written using only two digits rather than four to define the applicable year. Accordingly, the company's software may recognize a date using "00" as 1900 rather than the year 2000.

00



Duality of Erosion and Dilation

- Erosion and dilation are duals of each other with respect to set complementation and reflection.

$$(A \ominus B)^c = A^c \oplus \hat{B}$$

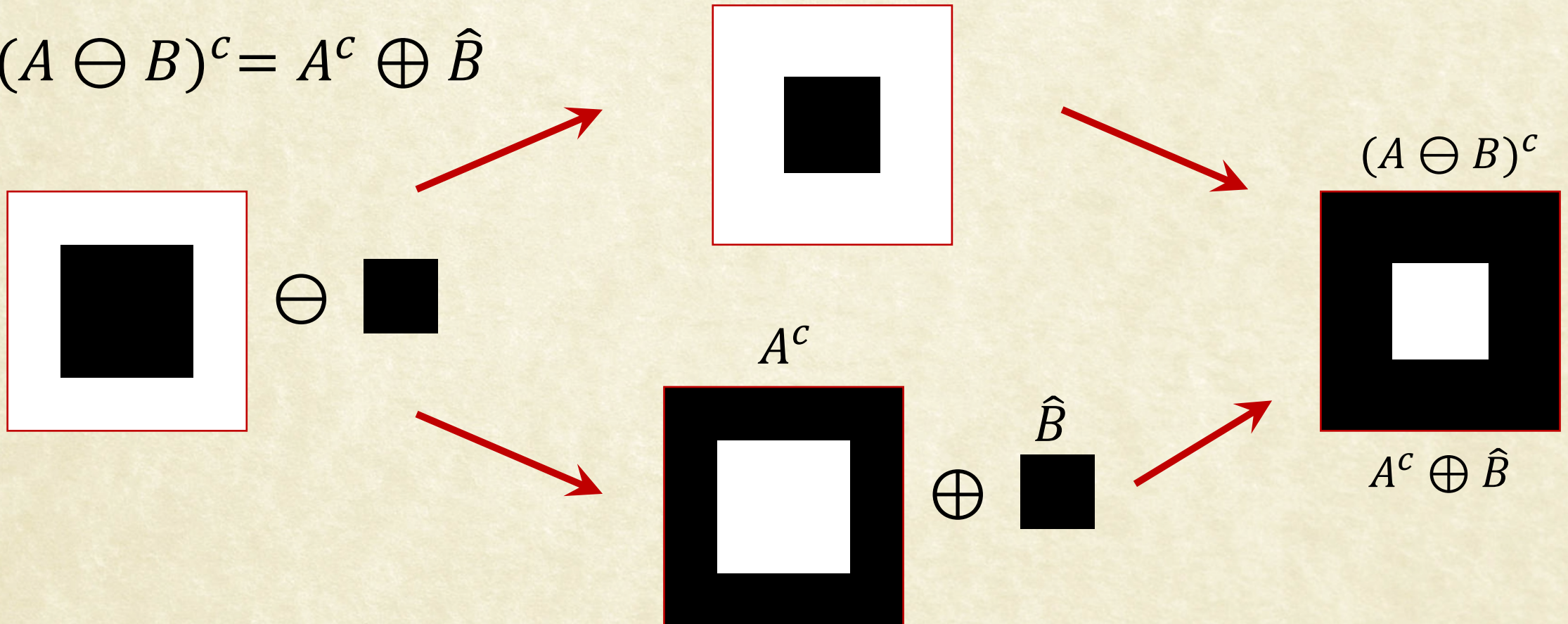
$$(A \oplus B)^c = A^c \ominus \hat{B}$$



Duality of Erosion and Dilation

- Erosion and dilation are duals of each other with respect to set complementation and reflection.

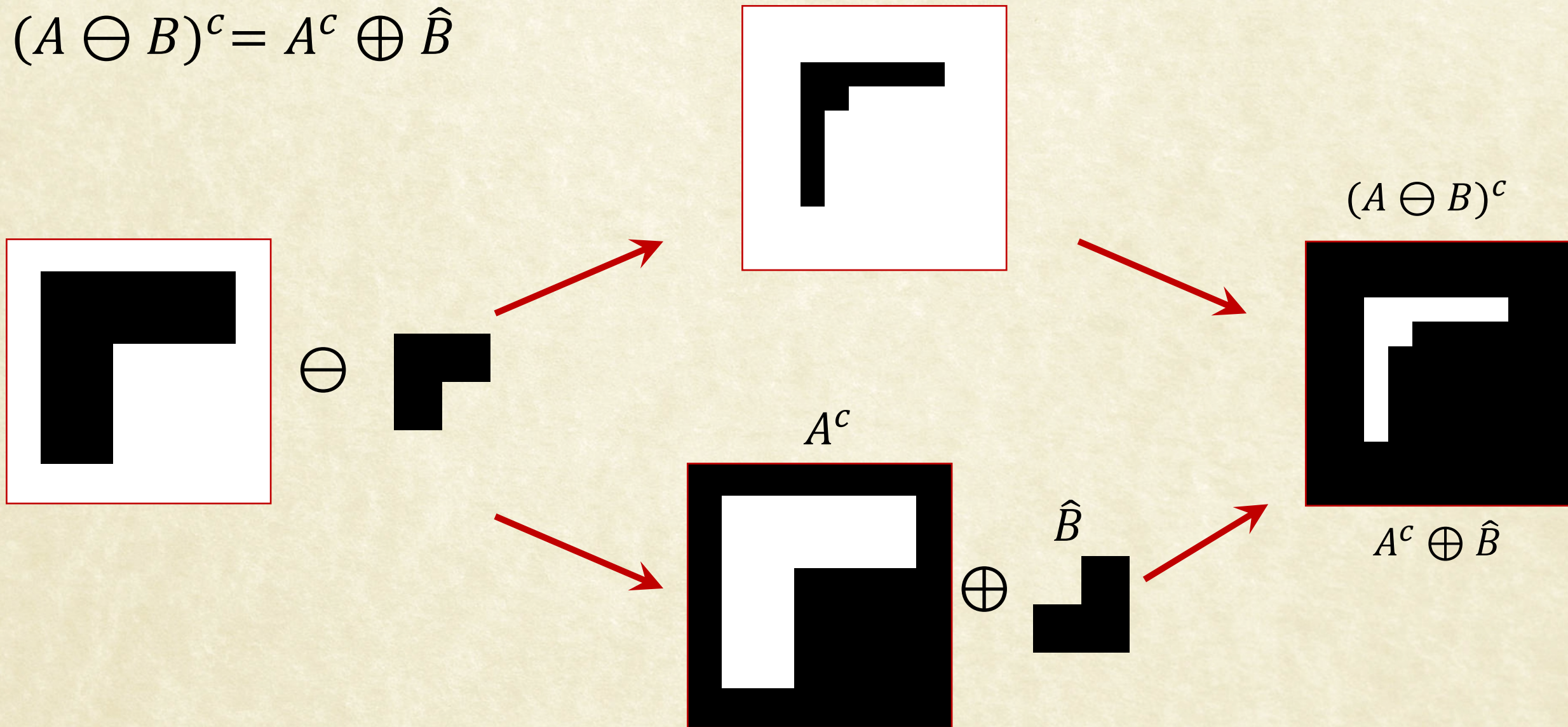
$$(A \ominus B)^c = A^c \oplus \hat{B}$$





Duality of Erosion and Dilation

$$(A \ominus B)^c = A^c \oplus \hat{B}$$





Duality: Proof

- By definition: $A \ominus B = \{z | (B)_z \subseteq A\} = \{z | (B)_z \cap A^c = \emptyset\}$
 - Therefore: $(A \ominus B)^c = (\{z | (B)_z \cap A^c = \emptyset\})^c$
 - Note: Complement of the set z satisfying $(B)_z \cap A^c = \emptyset$ is the set z that satisfies $(B)_z \cap A^c \neq \emptyset$.
 - Therefore: $(A \ominus B)^c = \{z | (B)_z \cap A^c \neq \emptyset\}$
 - i.e., $(A \ominus B)^c = A^c \oplus \hat{B}$ ■
- $A \oplus B = \{z | [(\hat{B})_z \cap A] \neq \emptyset\}$



Opening and Closing

- We would like to maintain the size of objects while filling holes or removing connections
- Opening: Erode, then Dilate

$$A \circ B = (A \ominus B) \oplus B$$

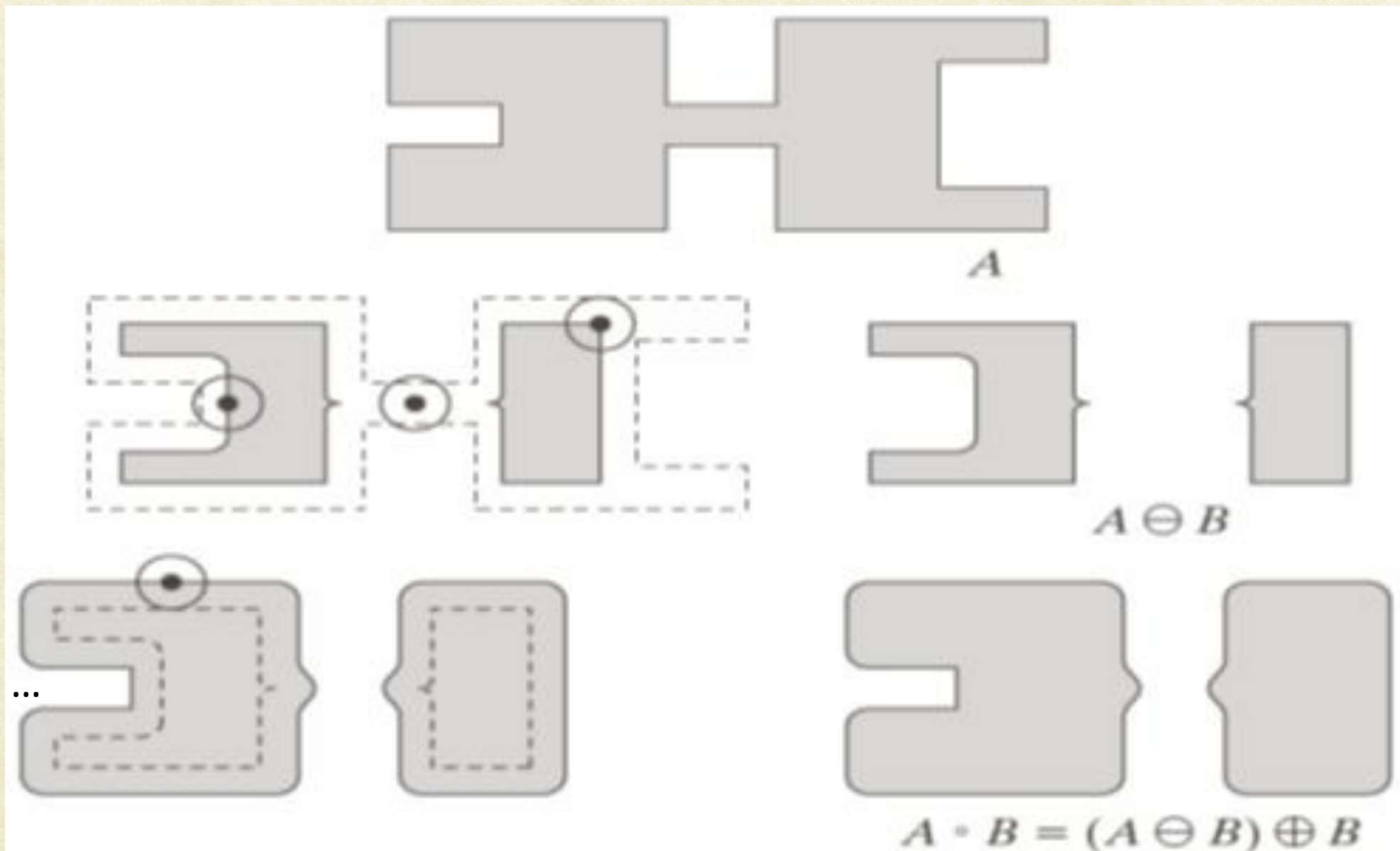
- Closing: Dilate, then Erode

$$A \cdot B = (A \oplus B) \ominus B$$



Opening:

Erosion

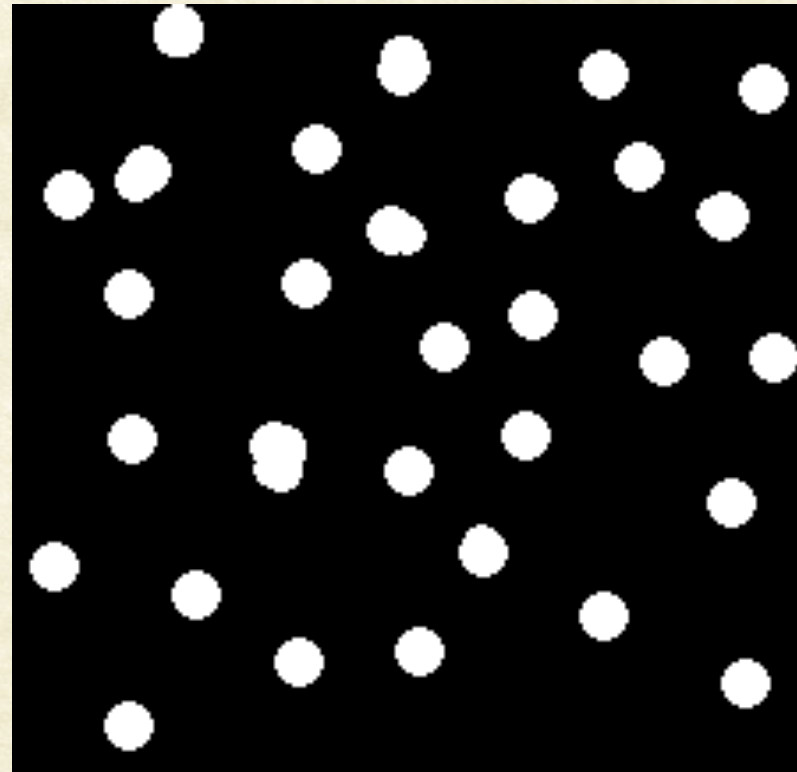
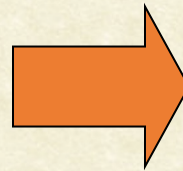
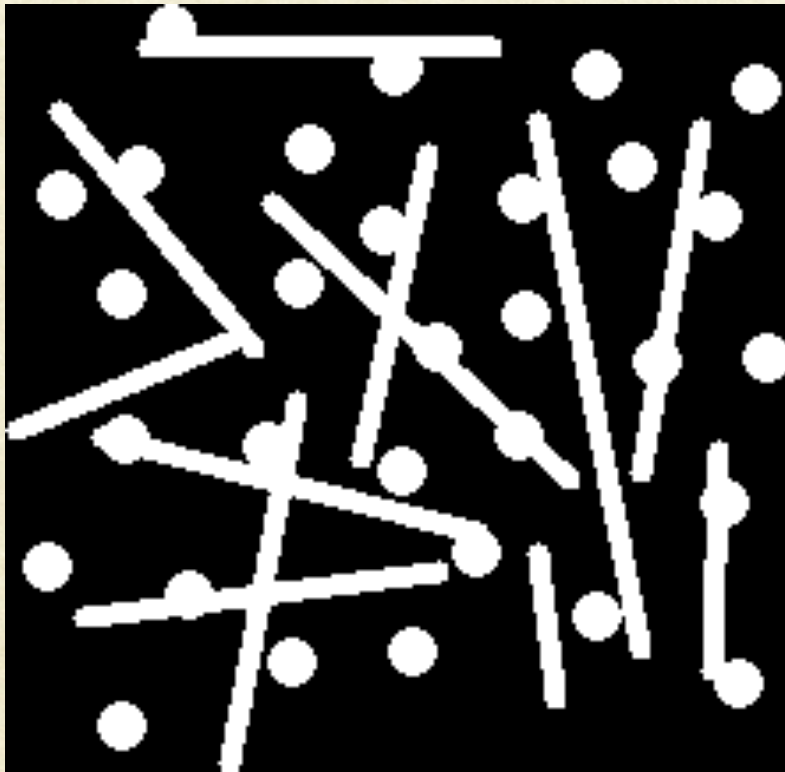


Followed by dilation ...



Opening: Example

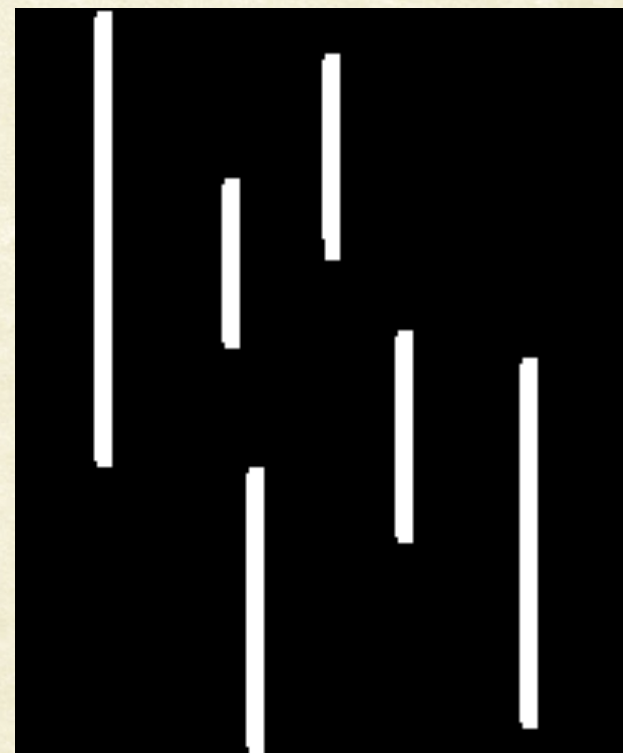
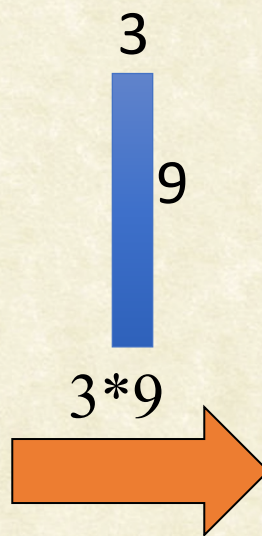
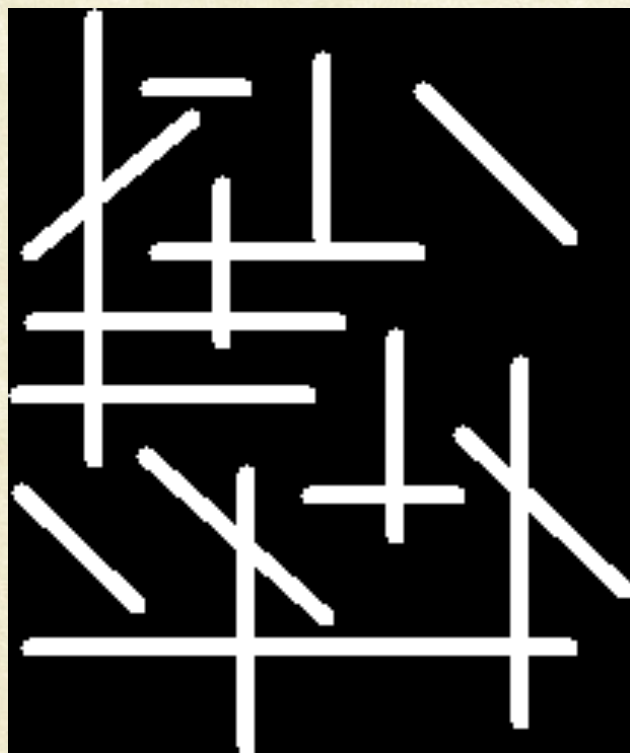
- Opening with a 11 pixel diameter disc





Opening: Another Example

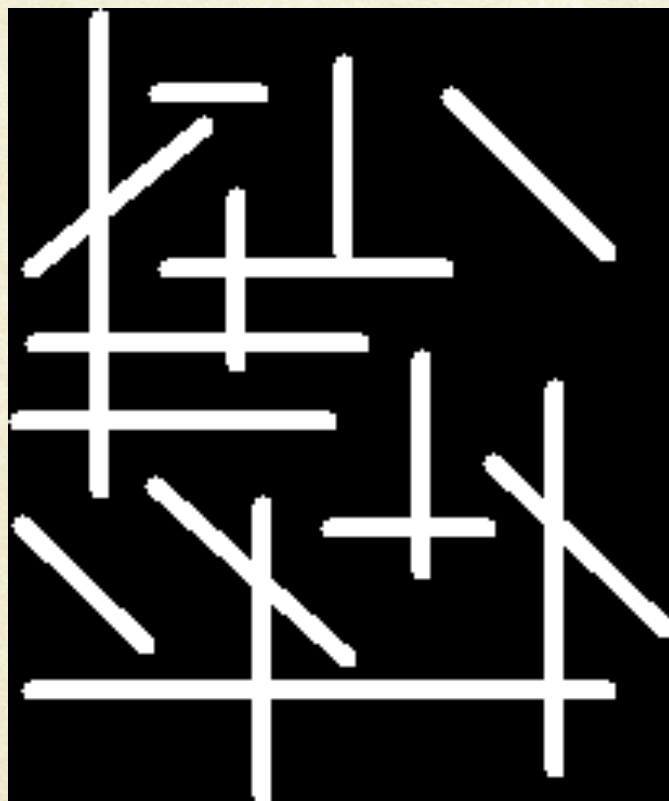
- 3x9 Structuring Element



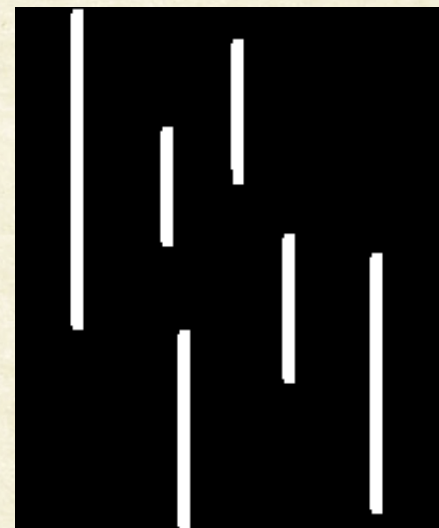
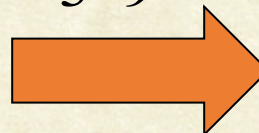


Opening: Another Example

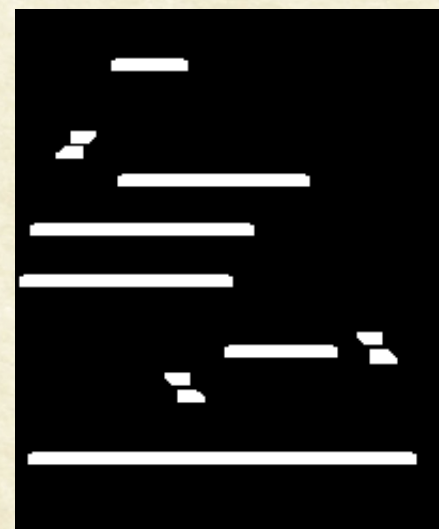
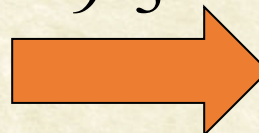
- 3x9 and 9x3 Structuring Element



3×9



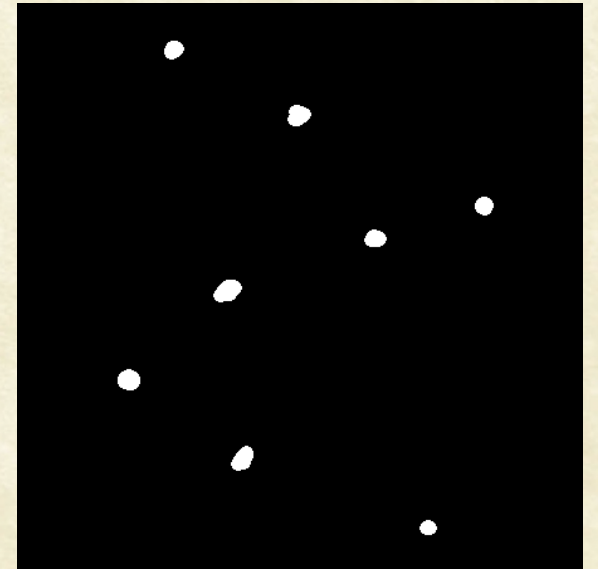
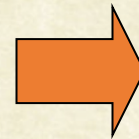
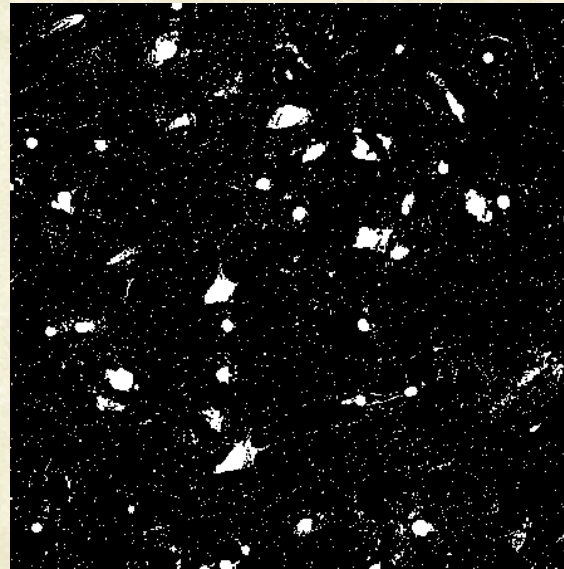
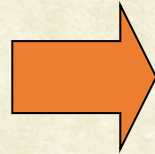
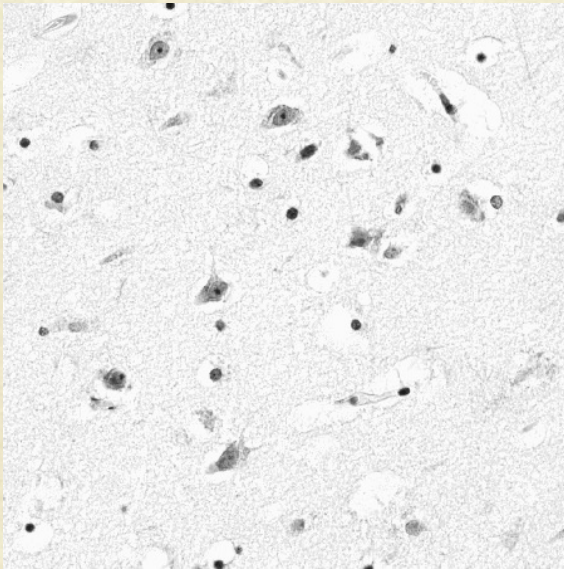
9×3





Use Opening for Separating Blobs

- Use large structuring element that fits into the big blobs
- Structuring Element: 11 pixel disc





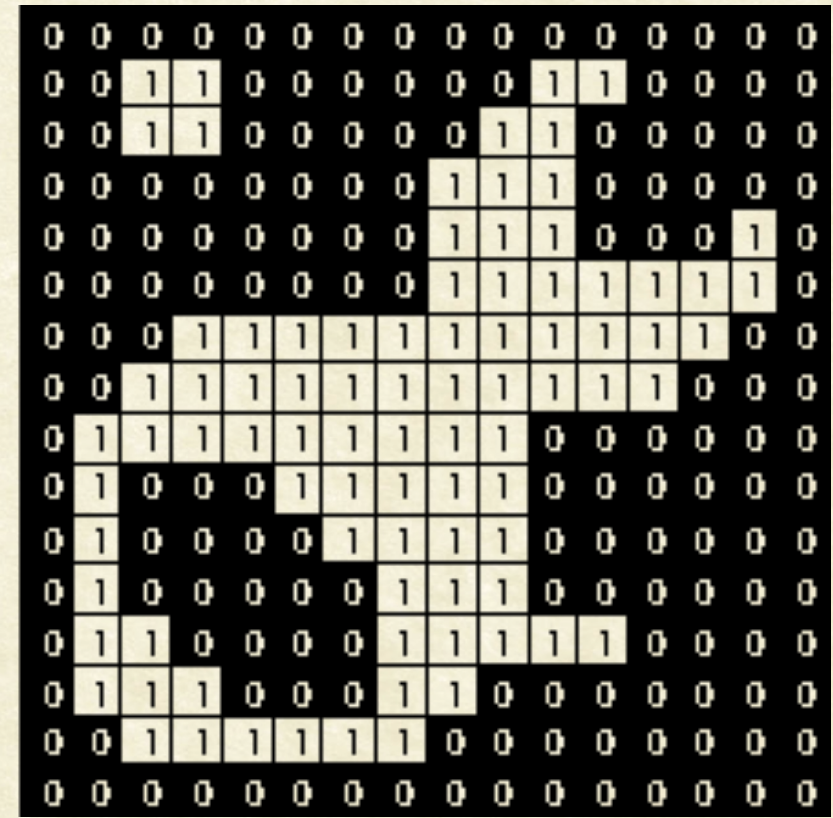
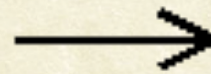
Opening

- Similar to Erosion
 - Spot and noise removal
 - Less destructive
- Erosion followed by Dilation
- *Same structuring element for both operations.*
- Opening is **idempotent**
 - Repeated application has no further effects!



Closing (Dilation then Erosion)

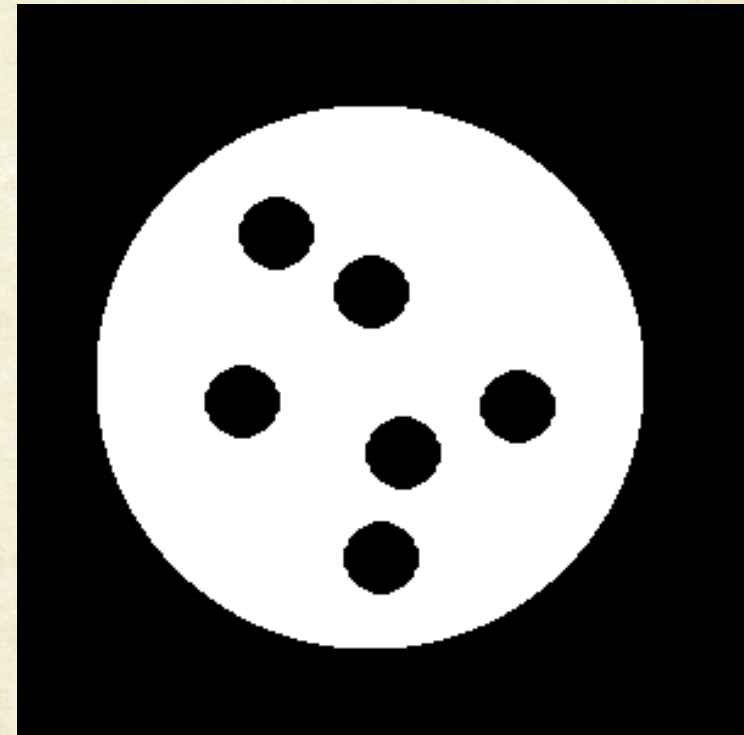
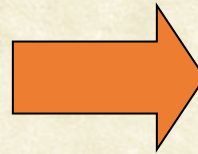
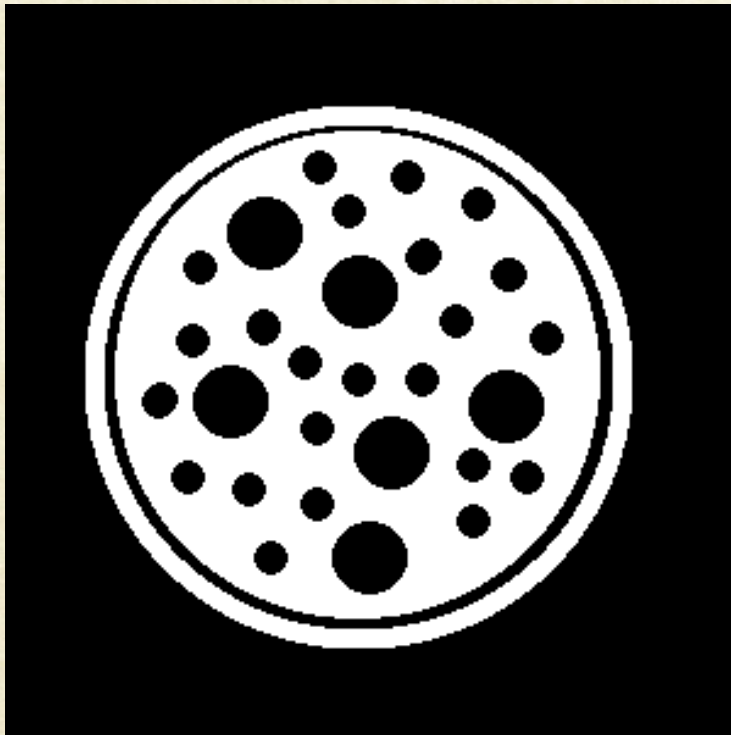
- Structuring element: 3x3 square





Closing: Example

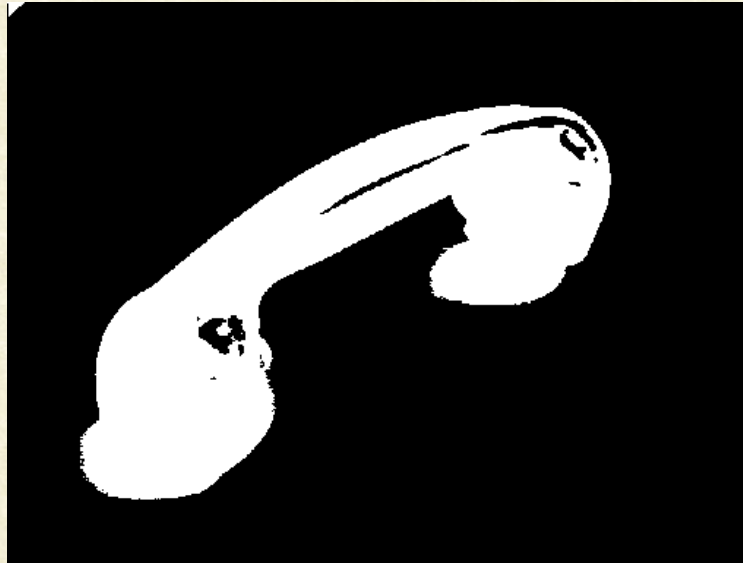
- Closing operation with a 22 pixel disc
- Closes small holes in the foreground



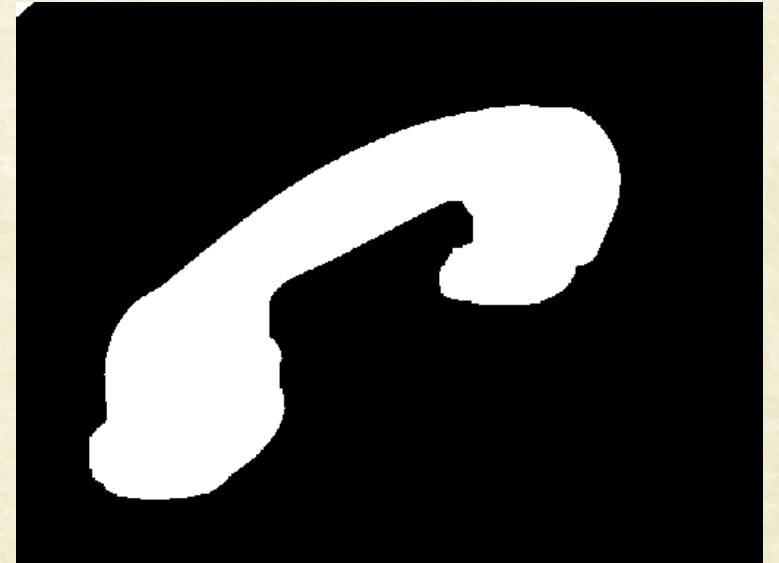


Closing Example

1. Threshold
2. Closing with disc of size 20



Thresholded

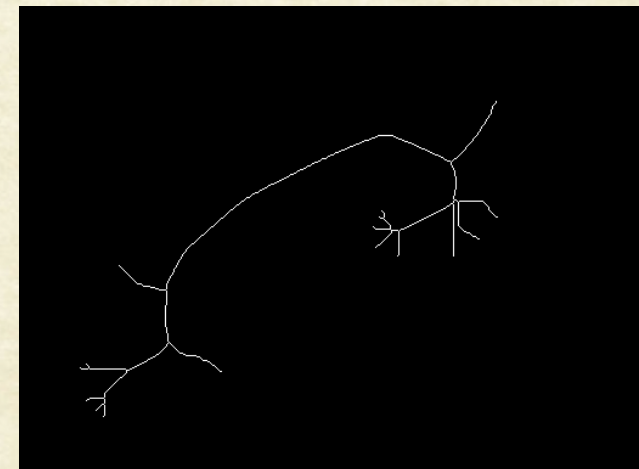
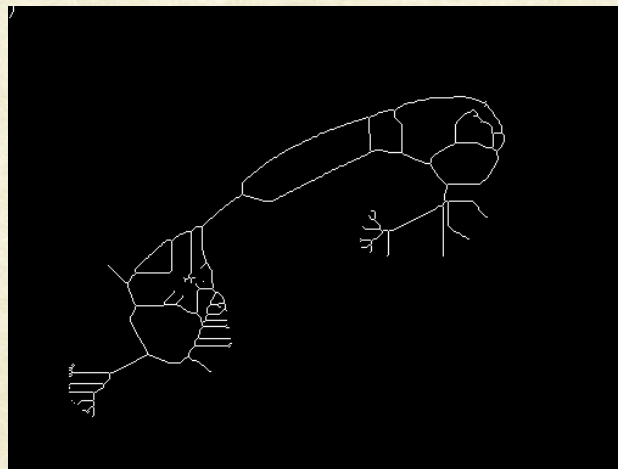
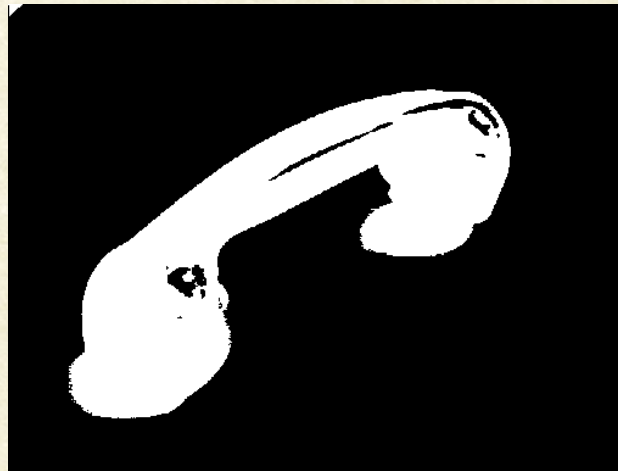


Closed



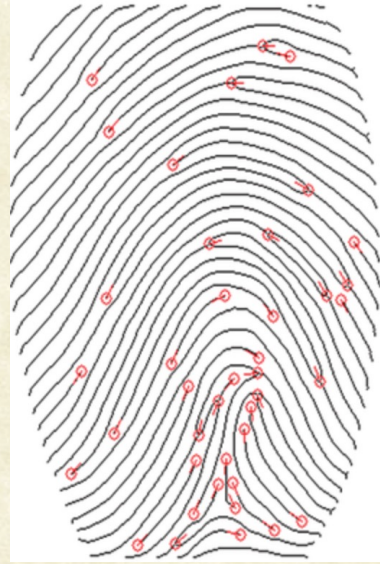
Application of Closing

- Good for further processing: E.g. Skeleton operation looks better for closed image!





Applications: Revisit



Minutiae
Extraction



Background
Subtraction



Questions?